

# DevOps Shack

# Linux

# Troubleshooting

# Guide

150 Common Errors — Root Cause Analysis & Solutions

2026 Edition

Comprehensive DevOps Reference Manual for Linux Engineers

**Classification: Internal Use Only**

Maintained by: DevOps Shack Engineering Team

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# How to Use This Guide

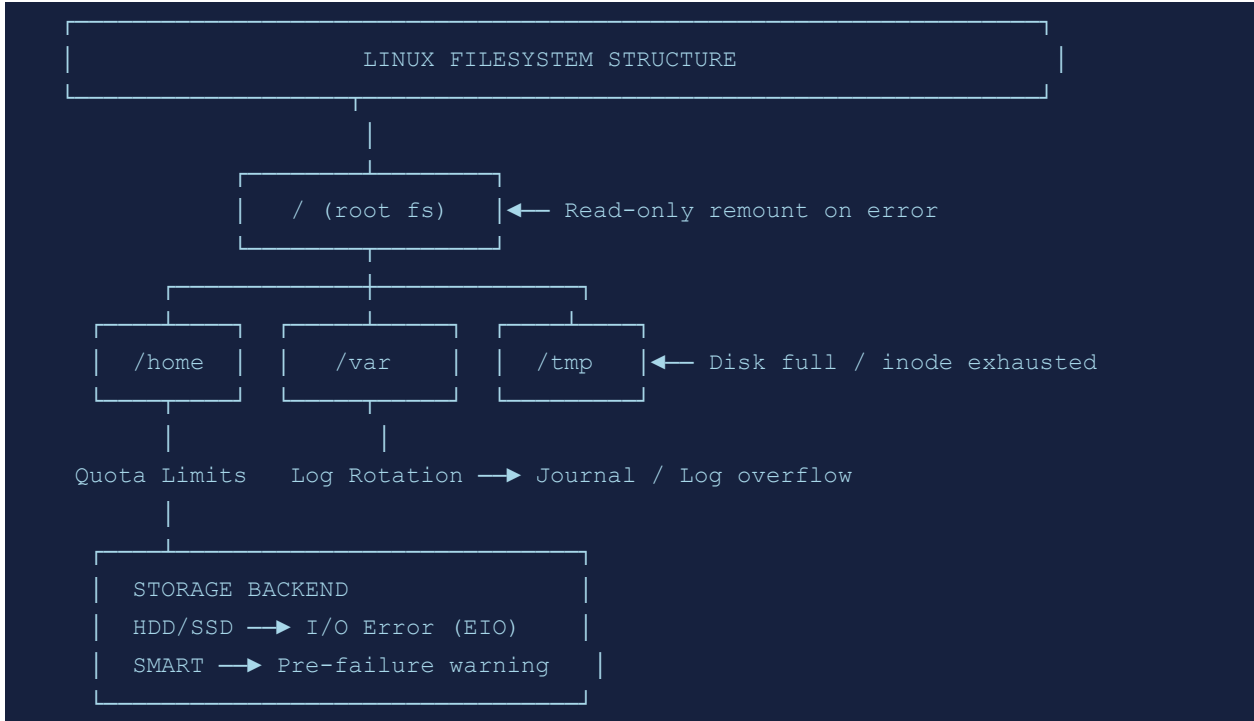
This guide covers 150 of the most common Linux errors encountered in production DevOps environments. Each entry is structured to help you quickly diagnose and resolve issues.

- **ERROR:** The exact error message or error code you will see in logs or terminal output.
- **DESCRIPTION:** What the error means in plain terms.
- **ROOT CAUSE:** The technical mechanism that triggers this error.
- **CAUSE:** The real-world scenario that leads to this error.
- **SOLUTION:** Step-by-step remediation actions ordered from safest/quickest to more drastic.

|                          |                        |
|--------------------------|------------------------|
| DESCRIPTION (Light Blue) | ROOT CAUSE (Soft Blue) |
| CAUSE (Warm Yellow)      | SOLUTION (Soft Green)  |

Section 1: Filesystem & Disk Errors (1–10)

Diagram: Linux Filesystem Hierarchy & Error Points



| #001 No space left on device |                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                  | The system cannot write to the disk because all available space has been consumed.                                                                                                                                                                                                                                                                                  |
| ROOT CAUSE                   | The filesystem reached 100% utilization, preventing any further writes including temp files, logs, or application data.                                                                                                                                                                                                                                             |
| CAUSE                        | Uncontrolled log growth, large temp files, or a runaway process filling the disk.                                                                                                                                                                                                                                                                                   |
| SOLUTION                     | <ul style="list-style-type: none"><li>Run <code>df -h</code> to identify the full partition</li><li>Use <code>du -sh /*</code> to find large directories</li><li>Remove old logs: <code>journalctl --vacuum-time=7d</code></li><li>Clear package cache: <code>apt clean</code> or <code>yum clean all</code></li><li>Archive or delete large unused files</li></ul> |

| #002 Read-only file system |                                                                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                | Attempts to write to the filesystem fail because it has been remounted read-only.                                                                     |
| ROOT CAUSE                 | The kernel detected filesystem corruption or disk errors and remounted it read-only to prevent further damage.                                        |
| CAUSE                      | Disk I/O errors, sudden power loss, or filesystem corruption.                                                                                         |
| SOLUTION                   | <ul style="list-style-type: none"><li>Check <code>dmesg</code> for hardware errors</li><li>Run <code>fsck /dev/sdXN</code> after unmounting</li></ul> |

- Check /etc/fstab for errors=remount-ro option
- Replace failing hardware if SMART errors exist: smartctl -a /dev/sdX

### #003 inode: No space left

|             |                                                                                                                                                                                                                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The filesystem has run out of inodes even though disk space may remain.                                                                                                                                                                                                                                                                                        |
| ROOT CAUSE  | Each file consumes one inode; millions of tiny files can exhaust the inode table before disk bytes are consumed.                                                                                                                                                                                                                                               |
| CAUSE       | Mail spools, session files, or cache directories accumulating millions of small files.                                                                                                                                                                                                                                                                         |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check with <code>df -i</code> to see inode usage</li><li>• Find directories with many files: <code>find /tmp -maxdepth 1 -type d   xargs -l{} sh -c "echo {} \$(ls {}   wc -l)"</code></li><li>• Clean mail queues, tmp dirs, or cache stores</li><li>• Consider reformatting with more inodes if persistent</li></ul> |

### #004 Permission denied

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A user or process attempted to access a file or directory without the required permissions.                                                                                                                                                                                                                                                                                                                                            |
| ROOT CAUSE  | UNIX permission model denied the action because the process UID/GID does not match the required permissions.                                                                                                                                                                                                                                                                                                                           |
| CAUSE       | Incorrect file permissions, wrong ownership, or SELinux/AppArmor policy denial.                                                                                                                                                                                                                                                                                                                                                        |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check permissions: <code>ls -la &lt;file&gt;</code></li><li>• Fix ownership: <code>chown user:group &lt;file&gt;</code></li><li>• Fix permissions: <code>chmod 644 &lt;file&gt;</code> or <code>chmod 755 &lt;dir&gt;</code></li><li>• Check SELinux denials: <code>audit2why &lt; /var/log/audit/audit.log</code></li><li>• Use <code>sudo</code> if elevated privileges are needed</li></ul> |

### #005 Too many open files

|             |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A process or the system has reached the limit of simultaneously open file descriptors.                                                                                                                                                                                                                                                                                                                                   |
| ROOT CAUSE  | The OS enforces limits on open file descriptors per process ( <code>ulimit</code> ) and system-wide. Reaching these limits causes failures.                                                                                                                                                                                                                                                                              |
| CAUSE       | File descriptor leak in application code, or limits too low for high-concurrency workloads.                                                                                                                                                                                                                                                                                                                              |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check limits: <code>ulimit -n</code> and <code>cat /proc/sys/fs/file-max</code></li><li>• Increase per-process limit: <code>ulimit -n 65536</code></li><li>• Set persistently in <code>/etc/security/limits.conf</code></li><li>• Increase system limit: <code>sysctl -w fs.file-max=200000</code></li><li>• Check leaks with <code>lsof -p &lt;PID&gt;   wc -l</code></li></ul> |

**#006 Disk quota exceeded**

|             |                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A user or group has reached their assigned disk quota and cannot write more data.                                                                                                                                                                                                                                                                                              |
| ROOT CAUSE  | The quota subsystem tracks disk usage per user/group and enforces hard limits set by the administrator.                                                                                                                                                                                                                                                                        |
| CAUSE       | User data growth exceeding configured quota limits.                                                                                                                                                                                                                                                                                                                            |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check quota: <code>quota -vs &lt;username&gt;</code></li><li>• View all quotas: <code>repquota -a</code></li><li>• Increase quota: <code>edquota -u &lt;username&gt;</code></li><li>• Help user clean up: <code>ncdu &lt;home_dir&gt;</code></li><li>• Disable quota temporarily if urgent: <code>quotaoff /partition</code></li></ul> |

**#007 File not found (ENOENT)**

|             |                                                                                                                                                                                                                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A process attempted to access a file or directory that does not exist at the given path.                                                                                                                                                                                                                                                 |
| ROOT CAUSE  | The VFS layer could not resolve the path to an existing inode in the filesystem.                                                                                                                                                                                                                                                         |
| CAUSE       | Wrong path, deleted file, unmounted filesystem, or broken symlink.                                                                                                                                                                                                                                                                       |
| SOLUTION    | <ul style="list-style-type: none"><li>• Verify path: <code>ls -la &lt;path&gt;</code></li><li>• Check symlinks: <code>readlink -f &lt;path&gt;</code></li><li>• Ensure filesystem is mounted: <code>mount   grep &lt;mount_point&gt;</code></li><li>• Check if file was deleted while in use: <code>lsif   grep deleted</code></li></ul> |

**#008 Input/output error**

|             |                                                                                                                                                                                                                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A low-level I/O error occurred when reading from or writing to a device.                                                                                                                                                                                                                                                                                 |
| ROOT CAUSE  | The storage hardware reported an error, causing the kernel to fail the I/O request.                                                                                                                                                                                                                                                                      |
| CAUSE       | Failing disk, bad sectors, faulty cables, or RAID degradation.                                                                                                                                                                                                                                                                                           |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check SMART data: <code>smartctl -H /dev/sdX</code></li><li>• Review kernel errors: <code>dmesg   grep -i "error\ ata\ sda"</code></li><li>• Run badblocks: <code>badblocks -sv /dev/sdX (read-only)</code></li><li>• Replace drive if errors persist</li><li>• Check cable connections and controller</li></ul> |

**#009 Filesystem corruption (fsck)**

|             |                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The filesystem has structural inconsistencies detected by fsck or during mount.                         |
| ROOT CAUSE  | Power loss during writes, kernel bug, or hardware failure left the filesystem in an inconsistent state. |
| CAUSE       | Abrupt shutdown without sync, bad blocks, or journaling failure.                                        |
| SOLUTION    | <ul style="list-style-type: none"><li>• Boot to recovery mode</li></ul>                                 |

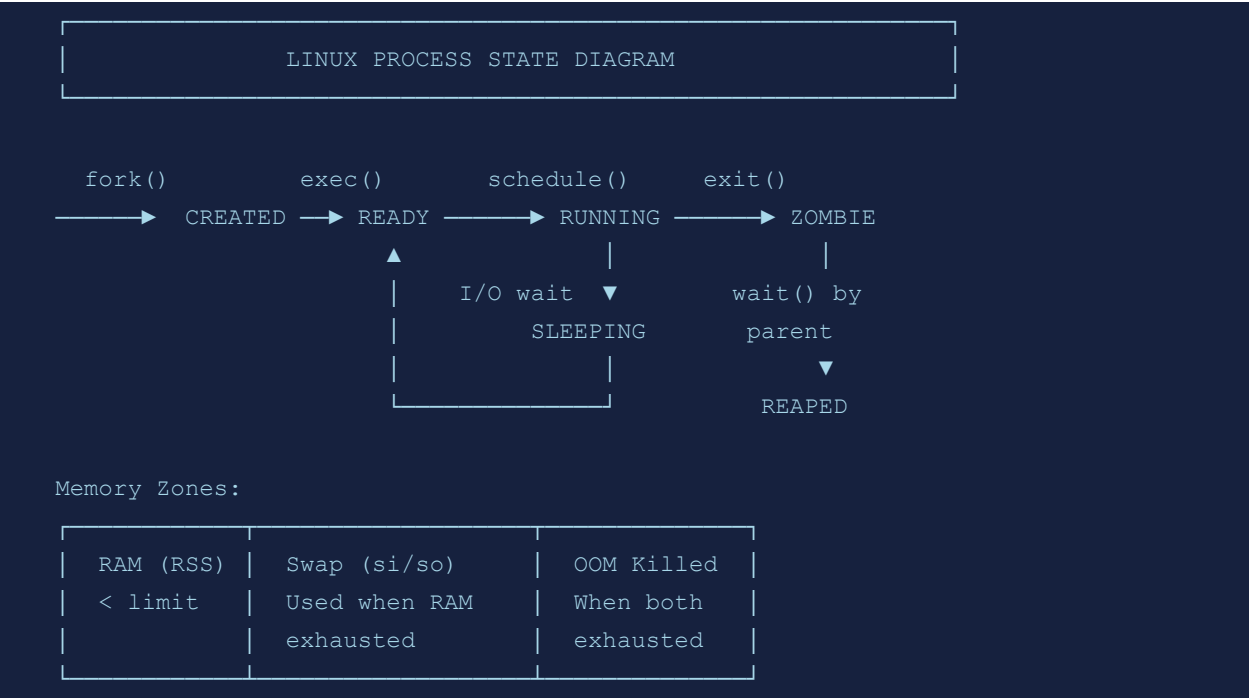
- Unmount the partition
- Run: `fsck -y /dev/sdXN`
- Review fsck output for repaired issues
- Restore from backup if fsck cannot repair

### #010 Mount: wrong filesystem type

|             |                                                                                                                                                                                                                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The mount command failed because the specified filesystem type does not match the device.                                                                                                                                                                                                            |
| ROOT CAUSE  | The kernel driver for the filesystem type could not interpret the device's on-disk format.                                                                                                                                                                                                           |
| CAUSE       | Wrong -t type specified, partition not formatted, or filesystem not supported by kernel.                                                                                                                                                                                                             |
| SOLUTION    | <ul style="list-style-type: none"><li>• Auto-detect type: <code>mount /dev/sdXN /mnt</code></li><li>• Inspect filesystem: <code>file -s /dev/sdXN</code></li><li>• Load kernel module: <code>modprobe &lt;fstype&gt;</code></li><li>• Format if new disk: <code>mkfs.ext4 /dev/sdXN</code></li></ul> |

## Section 2: Process & Memory Errors (11–20)

Diagram: Process Lifecycle & Memory States



### #011 Cannot allocate memory (ENOMEM)

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The kernel refused a memory allocation request because available memory is exhausted.                                                                                                                                                                                                                                                                                                                                                                 |
| ROOT CAUSE  | Physical RAM and swap space are both fully committed, leaving no headroom for additional allocations.                                                                                                                                                                                                                                                                                                                                                 |
| CAUSE       | Memory leak in application, insufficient swap, or overcommit limits reached.                                                                                                                                                                                                                                                                                                                                                                          |
| SOLUTION    | <ul style="list-style-type: none"><li>Check memory: <code>free -h</code> and <code>vmstat -s</code></li><li>Identify top consumers: <code>ps aux --sort=-%mem   head</code></li><li>Kill leaking process: <code>kill -9 &lt;PID&gt;</code></li><li>Add swap: <code>fallocate -l 4G /swapfile &amp;&amp; mkswap /swapfile &amp;&amp; swapon /swapfile</code></li><li>Tune <code>vm.overcommit_memory</code> in <code>/etc/sysctl.conf</code></li></ul> |

### #012 Segmentation fault (SIGSEGV)

|             |                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A process accessed memory outside its allowed address space and was terminated by the kernel.                      |
| ROOT CAUSE  | The CPU's MMU triggered a protection fault when the process attempted to read/write an unmapped or protected page. |
| CAUSE       | Bug in application code: null pointer dereference, buffer overflow, or use-after-free.                             |



|                 |                                                                                                                                                                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SOLUTION</b> | <ul style="list-style-type: none"><li>• Enable core dump: <code>ulimit -c unlimited</code></li><li>• Analyze with gdb: <code>gdb &lt;binary&gt; core</code></li><li>• Run under valgrind for memory errors</li><li>• Check for updates or patches to the software</li><li>• Review recent code changes</li></ul> |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### #013 OOM Killer: process killed

|                    |                                                                                                                                                                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | The Out-Of-Memory killer terminated a process to reclaim memory and prevent system panic.                                                                                                                                                                                                                           |
| <b>ROOT CAUSE</b>  | The kernel OOM killer activates when memory is critically low and selects a process to kill based on OOM score.                                                                                                                                                                                                     |
| <b>CAUSE</b>       | Memory overcommit, no swap, or a runaway memory-consuming process.                                                                                                                                                                                                                                                  |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Review: <code>dmesg   grep -i "oom"</code></li><li>• Identify which process was killed and why</li><li>• Protect critical services: <code>echo -17 &gt; /proc/&lt;PID&gt;/oom_adj</code></li><li>• Add more RAM or swap</li><li>• Fix memory leaks in application</li></ul> |

### #014 Process not found (no such process)

|                    |                                                                                                                                                                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | An attempt to signal or interact with a PID that does not exist in the process table.                                                                                                                                                                                              |
| <b>ROOT CAUSE</b>  | The target process has already exited, was never started, or the PID was recycled.                                                                                                                                                                                                 |
| <b>CAUSE</b>       | Race condition, script using stale PID file, or process crashed before signal was sent.                                                                                                                                                                                            |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Verify process: <code>ps aux   grep &lt;name&gt;</code></li><li>• Check PID files for staleness</li><li>• Use process name instead: <code>pkill -f &lt;name&gt;</code></li><li>• Implement proper PID file management in scripts</li></ul> |

### #015 Resource temporarily unavailable (EAGAIN)

|                    |                                                                                                                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A non-blocking I/O operation or resource acquisition failed because the resource is temporarily busy.                                                                                                                                                                                                          |
| <b>ROOT CAUSE</b>  | The kernel returned EAGAIN to indicate the caller should retry later, as the resource is not immediately available.                                                                                                                                                                                            |
| <b>CAUSE</b>       | File lock contention, pipe full, or socket buffer full in non-blocking mode.                                                                                                                                                                                                                                   |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Implement retry logic with exponential backoff</li><li>• Check file locks: <code>lsuf   grep LOCK</code></li><li>• Increase kernel socket buffer: <code>sysctl -w net.core.rmem_max=26214400</code></li><li>• Review application's non-blocking I/O handling</li></ul> |

**#016 Killed (signal 9)**

|             |                                                                                                                                                                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A process received SIGKILL and was immediately terminated without cleanup.                                                                                                                                                                           |
| ROOT CAUSE  | SIGKILL cannot be caught or ignored; it directly instructs the kernel to destroy the process.                                                                                                                                                        |
| CAUSE       | OOM killer, manual kill -9, or watchdog timeout.                                                                                                                                                                                                     |
| SOLUTION    | <ul style="list-style-type: none"><li>• Determine why kill -9 was needed vs graceful kill -15</li><li>• Check OOM events in dmesg</li><li>• Investigate why graceful shutdown failed</li><li>• Ensure application handles SIGTERM properly</li></ul> |

**#017 Zombie process**

|             |                                                                                                                                                                                                                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A dead process remains in the process table because its parent has not yet called wait() to collect its exit status.                                                                                                                                                                                                              |
| ROOT CAUSE  | The kernel retains process table entries for dead processes until the parent reads the exit status.                                                                                                                                                                                                                               |
| CAUSE       | Bug in parent process not handling SIGCHLD, or parent is also stuck.                                                                                                                                                                                                                                                              |
| SOLUTION    | <ul style="list-style-type: none"><li>• Identify zombies: ps aux   grep Z</li><li>• Find parent: ps -o ppid= -p &lt;zombie_pid&gt;</li><li>• Send SIGCHLD to parent: kill -CHLD &lt;parent_pid&gt;</li><li>• Kill parent if it does not handle children properly</li><li>• Fix application to call wait() on child exit</li></ul> |

**#018 CPU soft lockup detected**

|             |                                                                                                                                                                                                                                                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The kernel watchdog detected that a CPU has been spinning in kernel code for more than 20 seconds.                                                                                                                                                                                  |
| ROOT CAUSE  | A kernel thread or interrupt handler held a lock or spun in a tight loop, starving other work on that CPU.                                                                                                                                                                          |
| CAUSE       | Kernel bug, driver issue, or misconfigured interrupt affinity.                                                                                                                                                                                                                      |
| SOLUTION    | <ul style="list-style-type: none"><li>• Review: dmesg   grep "soft lockup"</li><li>• Update kernel to latest stable version</li><li>• Check for known driver bugs</li><li>• Reboot to clear (data may be at risk)</li><li>• File bug with kernel version and lockup trace</li></ul> |

**#019 Load average too high**

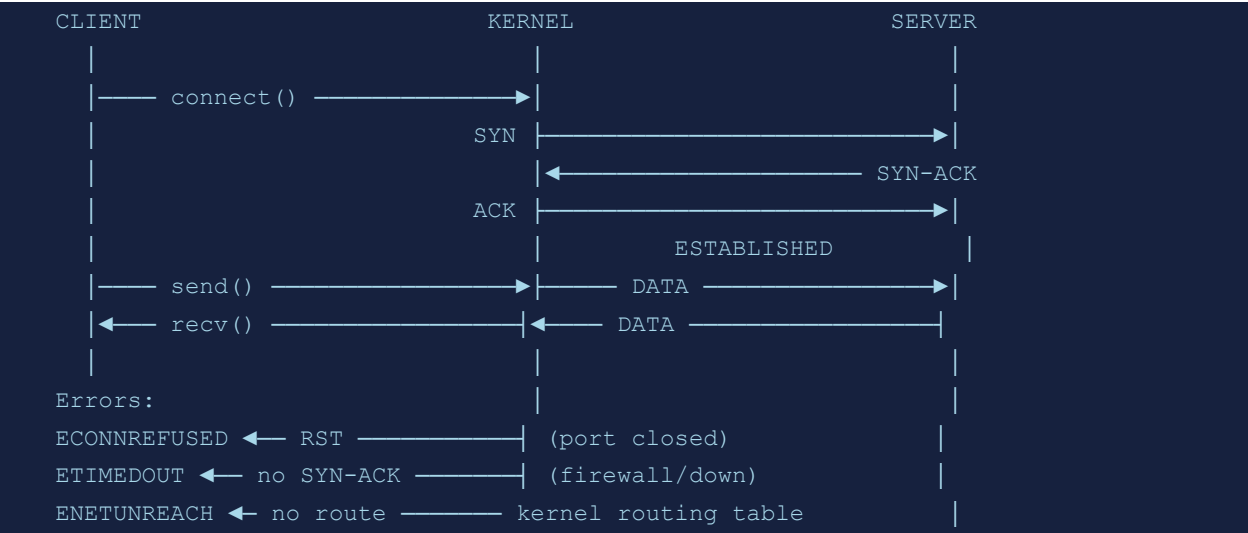
|             |                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | System load average significantly exceeds the number of CPU cores, indicating saturation.                   |
| ROOT CAUSE  | Load average counts processes in R (running) and D (uninterruptible sleep) state; high values mean queuing. |

|          |                                                                                                                                                                                                                                                                                  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAUSE    | CPU-bound workloads, I/O wait from slow storage, or process count explosion.                                                                                                                                                                                                     |
| SOLUTION | <ul style="list-style-type: none"><li>Inspect: top or htop</li><li>Check I/O wait: iostat -x 1</li><li>Identify D-state processes: ps aux   awk "\$8 == \"D\""</li><li>Scale horizontally or vertically</li><li>Tune I/O scheduler: cat /sys/block/sda/queue/scheduler</li></ul> |

|                                          |                                                                                                                                                                                                                                                       |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| #020 ulimit: max user processes exceeded |                                                                                                                                                                                                                                                       |
| DESCRIPTION                              | A new process cannot be forked because the user has reached their maximum process/thread count.                                                                                                                                                       |
| ROOT CAUSE                               | The OS limits the number of processes per user via nproc to prevent fork bombs and resource exhaustion.                                                                                                                                               |
| CAUSE                                    | Too many threads/processes spawned by services, or nproc limit too low.                                                                                                                                                                               |
| SOLUTION                                 | <ul style="list-style-type: none"><li>Check: ulimit -u</li><li>Increase in /etc/security/limits.conf: * soft nproc 65536</li><li>Restart affected services after adjusting limits</li><li>Check for runaway thread-spawning in applications</li></ul> |

Section 3: Networking Errors (21–30)

Diagram: TCP/IP Network Connection Lifecycle



#021 Connection refused (ECONNREFUSED)

|             |                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A TCP connection attempt was rejected by the remote host because no service is listening on the specified port.                                                                                                                                                                                                                                                                                 |
| ROOT CAUSE  | The remote kernel sent a TCP RST because the destination port has no socket in LISTEN state.                                                                                                                                                                                                                                                                                                    |
| CAUSE       | Service not started, wrong port, firewall RST rule, or service crashed.                                                                                                                                                                                                                                                                                                                         |
| SOLUTION    | <ul style="list-style-type: none"><li>Check service status: <code>systemctl status &lt;service&gt;</code></li><li>Verify listening ports: <code>ss -tlnp   grep &lt;port&gt;</code></li><li>Check firewall: <code>iptables -L</code> or <code>ufw status</code></li><li>Start service: <code>systemctl start &lt;service&gt;</code></li><li>Verify correct host/port in client config</li></ul> |

#022 Connection timed out (ETIMEDOUT)

|             |                                                                                                                                                                                                                                                                                                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A network connection attempt did not receive a response within the timeout period.                                                                                                                                                                                                                                                  |
| ROOT CAUSE  | TCP SYN packets were sent but no SYN-ACK was received, causing the connection to expire after retransmit attempts.                                                                                                                                                                                                                  |
| CAUSE       | Firewall silently dropping packets, incorrect routing, host down, or network congestion.                                                                                                                                                                                                                                            |
| SOLUTION    | <ul style="list-style-type: none"><li>Test reachability: <code>ping &lt;host&gt;</code></li><li>Trace route: <code>traceroute &lt;host&gt;</code></li><li>Check firewall rules on both ends</li><li>Verify routing table: <code>ip route</code></li><li>Use <code>tcpdump</code> to confirm SYN packets leaving interface</li></ul> |

**#023 Network is unreachable**

|                    |                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | The kernel has no route for the destination network in the routing table.                                                                                                                                                                                                                                                                                     |
| <b>ROOT CAUSE</b>  | When no route matches the destination, the kernel returns ENETUNREACH to the application.                                                                                                                                                                                                                                                                     |
| <b>CAUSE</b>       | Missing default gateway, downed network interface, or misconfigured routing table.                                                                                                                                                                                                                                                                            |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Check routes: <code>ip route show</code></li><li>• Add default gateway: <code>ip route add default via &lt;gateway_ip&gt;</code></li><li>• Bring up interface: <code>ip link set eth0 up</code></li><li>• Verify network config files in <code>/etc/netplan/</code> or <code>/etc/network/interfaces</code></li></ul> |

**#024 DNS resolution failure**

|                    |                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A hostname cannot be resolved to an IP address, blocking network connections.                                                                                                                                                                                                                                                                                                       |
| <b>ROOT CAUSE</b>  | The resolver library (glibc) could not obtain a valid DNS answer from any configured nameserver.                                                                                                                                                                                                                                                                                    |
| <b>CAUSE</b>       | Wrong DNS server in <code>/etc/resolv.conf</code> , DNS server down, NXDOMAIN, or network outage.                                                                                                                                                                                                                                                                                   |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Test: <code>dig @8.8.8.8 &lt;hostname&gt;</code></li><li>• Check <code>/etc/resolv.conf</code> for valid nameservers</li><li>• Verify DNS service: <code>systemctl status systemd-resolved</code></li><li>• Flush DNS cache: <code>resolvectl flush-caches</code></li><li>• Check <code>/etc/nsswitch.conf</code> for hosts order</li></ul> |

**#025 SSH: Connection reset by peer**

|                    |                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | An established SSH connection was unexpectedly terminated by the remote side.                                                                                                                                                                                                                                                                                    |
| <b>ROOT CAUSE</b>  | The TCP connection was closed with RST rather than FIN, indicating an abrupt termination.                                                                                                                                                                                                                                                                        |
| <b>CAUSE</b>       | SSH server crash, firewall idle timeout, network interruption, or fail2ban blocking.                                                                                                                                                                                                                                                                             |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Check sshd status: <code>systemctl status sshd</code></li><li>• Review <code>/var/log/auth.log</code> for errors</li><li>• Set <code>ClientAliveInterval</code> in <code>sshd_config</code></li><li>• Check fail2ban: <code>fail2ban-client status sshd</code></li><li>• Verify firewall idle timeout settings</li></ul> |

**#026 Address already in use (EADDRINUSE)**

|                    |                                                                                          |
|--------------------|------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A service cannot bind to a network address/port because another process already owns it. |
| <b>ROOT CAUSE</b>  | TCP/UDP socket bind fails if the exact address:port combination is already in use.       |

|          |                                                                                                                                                                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAUSE    | Previous instance of service not fully stopped, TIME_WAIT state, or port conflict.                                                                                                                                                                                                                                                                        |
| SOLUTION | <ul style="list-style-type: none"><li>Find owner: <code>ss -tlnp   grep &lt;port&gt;</code> or <code>lsof -i :&lt;port&gt;</code></li><li>Kill conflicting process</li><li>Set <code>SO_REUSEADDR</code> in application</li><li>Wait for <code>TIME_WAIT</code> to expire (typically 60s)</li><li>Change service port if conflict is persistent</li></ul> |

## #027 iptables: no chain/target/match

|             |                                                                                                                                                                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | An iptables command references a chain, target, or match module that does not exist.                                                                                                                                                                              |
| ROOT CAUSE  | iptables requires specific kernel modules to be loaded for each match extension or custom chain to exist.                                                                                                                                                         |
| CAUSE       | Typo in chain name, kernel module not loaded, or missing iptables extension.                                                                                                                                                                                      |
| SOLUTION    | <ul style="list-style-type: none"><li>List chains: <code>iptables -L -n</code></li><li>Load module: <code>modprobe xt_recent</code></li><li>Check module availability: <code>lsmod   grep xt_</code></li><li>Verify command syntax in iptables man page</li></ul> |

## #028 Broken pipe (EPIPE)

|             |                                                                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A write to a pipe or socket failed because the reading end has been closed.                                                                                                                                                                                                                      |
| ROOT CAUSE  | The kernel delivers SIGPIPE (or returns EPIPE) when a process writes to a pipe with no readers.                                                                                                                                                                                                  |
| CAUSE       | Pipeline commands where a downstream process exits before upstream finishes writing.                                                                                                                                                                                                             |
| SOLUTION    | <ul style="list-style-type: none"><li>Ignore SIGPIPE in long-running services: <code>signal(SIGPIPE, SIG_IGN)</code></li><li>Check that downstream pipeline commands are running</li><li>Handle EPIPE return code gracefully in code</li><li>Review log for which process exited early</li></ul> |

## #029 Network interface not found

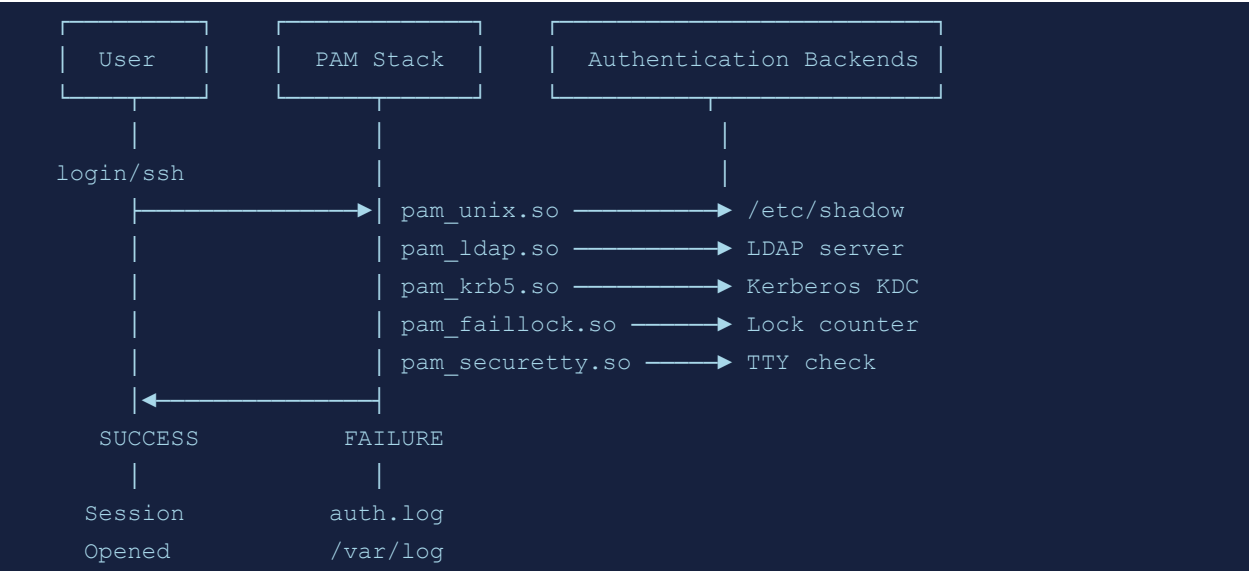
|             |                                                                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A network management command references an interface name that does not exist.                                                                                                                                                                                                                                      |
| ROOT CAUSE  | The kernel's network subsystem has no registered interface with the given name.                                                                                                                                                                                                                                     |
| CAUSE       | Interface renamed by udev, driver not loaded, or VM NIC not attached.                                                                                                                                                                                                                                               |
| SOLUTION    | <ul style="list-style-type: none"><li>List interfaces: <code>ip link show</code></li><li>Check driver: <code>lspci   grep Ethernet</code> then <code>modprobe &lt;driver&gt;</code></li><li>Check udev naming rules in <code>/etc/udev/rules.d/</code></li><li>For VMs, verify NIC attached in hypervisor</li></ul> |

**#030 SSL certificate verify failed**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A TLS/SSL handshake failed because the server's certificate could not be verified against trusted CAs.                                                                                                                                                                                                                                                                                                               |
| <b>ROOT CAUSE</b>  | The TLS library could not build a valid chain from the server cert to a trusted root CA in the local trust store.                                                                                                                                                                                                                                                                                                    |
| <b>CAUSE</b>       | Expired certificate, self-signed cert not trusted, wrong hostname, or outdated CA bundle.                                                                                                                                                                                                                                                                                                                            |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Check cert: <code>openssl s_client -connect host:443</code></li><li>• Update CA bundle: <code>update-ca-certificates</code></li><li>• Add custom CA: <code>cp ca.crt /usr/local/share/ca-certificates/ &amp;&amp; update-ca-certificates</code></li><li>• Check system date is correct: <code>timedatectl</code></li><li>• Verify certificate hostname matches SNI</li></ul> |

Section 4: Authentication & User Errors (31–40)

Diagram: Linux Authentication Flow (PAM)



| #031 Authentication failure |                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                 | A login attempt failed because the provided credentials do not match stored records.                                                                                                                                                                                                                                                                        |
| ROOT CAUSE                  | PAM compared the provided password hash against /etc/shadow and found no match, denying access.                                                                                                                                                                                                                                                             |
| CAUSE                       | Wrong password, account locked, expired password, or PAM misconfiguration.                                                                                                                                                                                                                                                                                  |
| SOLUTION                    | <ul style="list-style-type: none"><li>Reset password: <code>passwd &lt;username&gt;</code></li><li>Check account status: <code>chage -l &lt;username&gt;</code></li><li>Unlock account: <code>usermod -U &lt;username&gt;</code></li><li>Review PAM config: <code>/etc/pam.d/sshd</code></li><li>Check <code>/var/log/auth.log</code> for details</li></ul> |

| #032 sudo: user not in sudoers |                                                                                                                                                                                                                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                    | A user attempted to run sudo but is not authorized in the sudoers configuration.                                                                                                                                                                                                                |
| ROOT CAUSE                     | sudo checks /etc/sudoers and drop-in files; if the user is not listed, the command is denied and logged.                                                                                                                                                                                        |
| CAUSE                          | User was not added to sudo group or sudoers file.                                                                                                                                                                                                                                               |
| SOLUTION                       | <ul style="list-style-type: none"><li>Add to sudo group (as root): <code>usermod -aG sudo &lt;username&gt;</code></li><li>Or edit sudoers safely: <code>visudo</code></li><li>Add entry: <code>username ALL=(ALL:ALL) ALL</code></li><li>Log off and back on for group to take effect</li></ul> |



**#033 Account is locked**

|                    |                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | Login attempts fail because the account has been administratively locked or auto-locked after failed attempts.                                                                                                                                                                                                                                           |
| <b>ROOT CAUSE</b>  | PAM pam_tally2/pam_faillock increments failure counters and locks accounts after threshold is exceeded.                                                                                                                                                                                                                                                  |
| <b>CAUSE</b>       | Too many failed login attempts, or administrator manually locked the account.                                                                                                                                                                                                                                                                            |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Unlock with pam_tally2: <code>pam_tally2 --reset --user &lt;username&gt;</code></li><li>• Or pam_faillock: <code>faillock --reset --user &lt;username&gt;</code></li><li>• Or usermod: <code>usermod -U &lt;username&gt;</code></li><li>• Review <code>/var/log/auth.log</code> for source of failures</li></ul> |

**#034 SSH: Too many authentication failures**

|                    |                                                                                                                                                                                                                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | SSH client disconnects after trying too many keys, triggering the server's MaxAuthTries limit.                                                                                                                                                                                                                                       |
| <b>ROOT CAUSE</b>  | sshd counts each authentication method attempt; exceeding MaxAuthTries causes disconnection.                                                                                                                                                                                                                                         |
| <b>CAUSE</b>       | SSH agent has too many keys loaded, or client retries with multiple methods before password.                                                                                                                                                                                                                                         |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Specify key: <code>ssh -i ~/.ssh/specific_key user@host</code></li><li>• Limit agent keys: <code>ssh-add -D</code> then add only needed key</li><li>• Use IdentitiesOnly yes in <code>~/.ssh/config</code></li><li>• Increase MaxAuthTries in <code>sshd_config</code> (temporary)</li></ul> |

**#035 su: must be run from a terminal**

|                    |                                                                                                                                                                                                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | The su command was invoked without a controlling terminal, which PAM requires for interactive authentication.                                                                                                                                                                                        |
| <b>ROOT CAUSE</b>  | PAM's pam_securetty module or tty check requires a proper TTY for the su authentication flow.                                                                                                                                                                                                        |
| <b>CAUSE</b>       | Running su inside scripts, cron jobs, or non-interactive shells without TTY allocation.                                                                                                                                                                                                              |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Use <code>sudo -i</code> instead of <code>su</code></li><li>• Allocate PTY in SSH: <code>ssh -t user@host su -</code></li><li>• Use PAM configuration that allows non-TTY su</li><li>• Configure passwordless sudo for the specific command needed</li></ul> |

**#036 Group does not exist**

|                    |                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A command references a group name that is not present in <code>/etc/group</code> .                                |
| <b>ROOT CAUSE</b>  | The group database ( <code>/etc/group</code> and optionally LDAP/NIS) has no entry matching the given group name. |

|          |                                                                                                                                                                                                                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAUSE    | Typo in group name, group was deleted, or LDAP connectivity issue.                                                                                                                                                                                                                                     |
| SOLUTION | <ul style="list-style-type: none"><li>List groups: <code>getent group   grep &lt;groupname&gt;</code></li><li>Create group: <code>groupadd &lt;groupname&gt;</code></li><li>Check LDAP connectivity if using directory services</li><li>Verify <code>nsswitch.conf</code> group lookup order</li></ul> |

### #037 User does not exist

|             |                                                                                                                                                                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A command references a username that has no entry in the user database.                                                                                                                                                                                                                               |
| ROOT CAUSE  | The user database ( <code>/etc/passwd</code> or directory service) returns no result for the given username.                                                                                                                                                                                          |
| CAUSE       | User not created, typo, deleted account, or LDAP unreachable.                                                                                                                                                                                                                                         |
| SOLUTION    | <ul style="list-style-type: none"><li>Check: <code>id &lt;username&gt;</code> or <code>getent passwd &lt;username&gt;</code></li><li>Create user: <code>useradd -m &lt;username&gt;</code></li><li>Check LDAP/SSSD connectivity</li><li>Verify <code>/etc/nsswitch.conf</code> passwd entry</li></ul> |

### #038 PAM: system error

|             |                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A PAM module encountered an unexpected system-level error during authentication.                                                                                                                                                                                                                                                                                        |
| ROOT CAUSE  | A PAM module failed due to missing library, misconfiguration, or system call failure.                                                                                                                                                                                                                                                                                   |
| CAUSE       | Corrupted PAM config, missing PAM module file, or broken library dependencies.                                                                                                                                                                                                                                                                                          |
| SOLUTION    | <ul style="list-style-type: none"><li>Review <code>/etc/pam.d/</code> for syntax errors</li><li>Check module exists: <code>ls /lib/security/pam_*.so</code></li><li>Run: <code>pamtester &lt;service&gt; &lt;user&gt; authenticate</code></li><li>Review <code>/var/log/auth.log</code> for specific error</li><li>Reinstall pam package if module is missing</li></ul> |

### #039 Kerberos: kinit failed

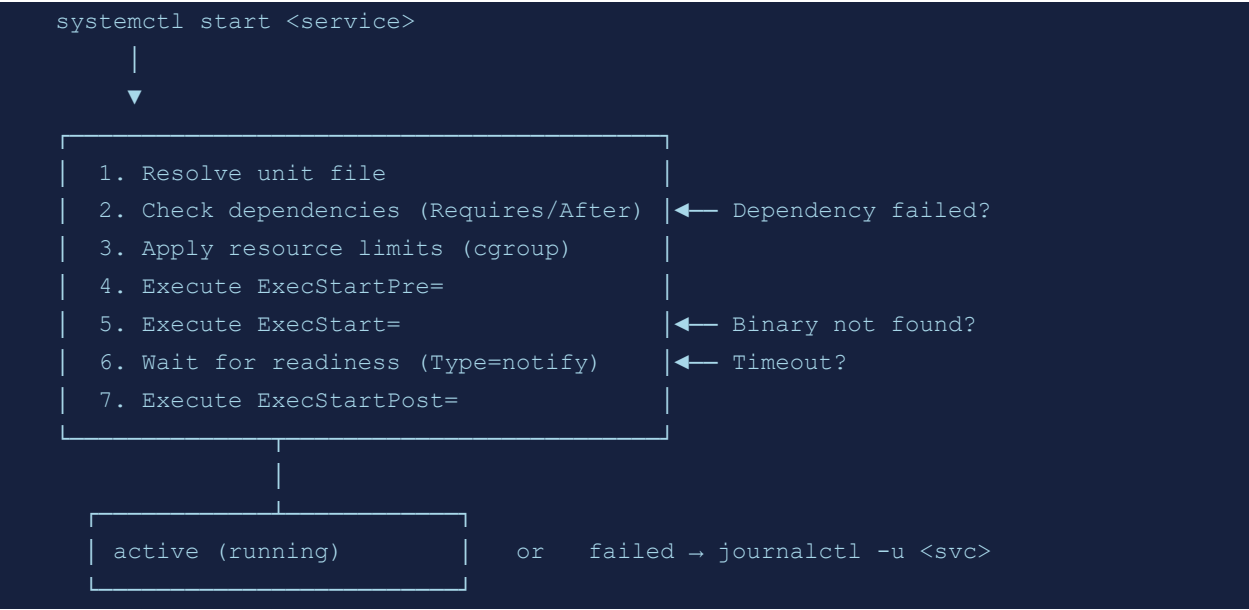
|             |                                                                                                                                                                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | Kerberos ticket initialization failed, preventing SSO authentication.                                                                                                                                                                                                                                            |
| ROOT CAUSE  | The KDC rejected the authentication request or the client configuration is incorrect.                                                                                                                                                                                                                            |
| CAUSE       | Wrong password, clock skew >5 minutes, unreachable KDC, or wrong realm.                                                                                                                                                                                                                                          |
| SOLUTION    | <ul style="list-style-type: none"><li>Check time sync: <code>timedatectl</code> and <code>chronyc</code> tracking</li><li>Verify KDC reachability: <code>nc -zv &lt;kdc_host&gt; 88</code></li><li>Check <code>/etc/krb5.conf</code> realm and KDC settings</li><li>Verify username and realm spelling</li></ul> |

**#040 LDAP: Can't contact LDAP server**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | The LDAP client cannot establish a connection to the configured directory server.                                                                                                                                                                                                                                                                                                                    |
| <b>ROOT CAUSE</b>  | The TCP connection to the LDAP port (389 or 636) failed or timed out.                                                                                                                                                                                                                                                                                                                                |
| <b>CAUSE</b>       | LDAP server down, wrong URI, firewall blocking, or TLS misconfiguration.                                                                                                                                                                                                                                                                                                                             |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Test: <code>ldapsearch -x -H ldap://&lt;server&gt; -b dc=example,dc=com</code></li><li>• Check connectivity: <code>nc -zv &lt;ldap_server&gt; 389</code></li><li>• Verify <code>/etc/ldap/ldap.conf</code> or <code>sssd.conf</code> settings</li><li>• Check TLS certs if using LDAPS</li><li>• Restart sssd: <code>systemctl restart sssd</code></li></ul> |

Section 5: Systemd & Service Errors (41–50)

Diagram: systemd Service Start Flow



#041 systemd: Unit failed to start

|             |                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A systemd service unit entered the failed state and could not start.                                                                                                                                                                                                                                                                                                            |
| ROOT CAUSE  | The service's main process exited with a non-zero code or could not be executed.                                                                                                                                                                                                                                                                                                |
| CAUSE       | Missing executable, configuration error, dependency not met, or resource limit hit.                                                                                                                                                                                                                                                                                             |
| SOLUTION    | <ul style="list-style-type: none"><li>Check status: <code>systemctl status &lt;service&gt;</code></li><li>View logs: <code>journalctl -u &lt;service&gt; -n 50</code></li><li>Test config: <code>&lt;binary&gt; --configtest</code></li><li>Check dependencies: <code>systemctl list-dependencies &lt;service&gt;</code></li><li>Run ExecStart manually to see errors</li></ul> |

#042 Failed to enable unit: File exists

|             |                                                                                                                                                                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | <code>systemctl enable</code> failed because the symlink target already exists.                                                                                                                                                                                                                                |
| ROOT CAUSE  | An existing file or symlink at the expected symlink path conflicts with the enable operation.                                                                                                                                                                                                                  |
| CAUSE       | Leftover symlink from previous install, or unit enabled through alternate method.                                                                                                                                                                                                                              |
| SOLUTION    | <ul style="list-style-type: none"><li>Check: <code>ls -la /etc/systemd/system/multi-user.target.wants/   grep &lt;unit&gt;</code></li><li>Remove stale symlink manually</li><li>Use <code>systemctl --force enable &lt;unit&gt;</code></li><li>Run <code>systemctl daemon-reload</code> after fixing</li></ul> |

#043 Job for service failed because a timeout was exceeded

|                    |                                                                                                                                                                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | systemd killed a service start/stop operation because it did not complete within the configured timeout.                                                                                                                                                                                   |
| <b>ROOT CAUSE</b>  | systemd sends SIGTERM then SIGKILL if a service does not transition state within TimeoutStartSec or TimeoutStopSec.                                                                                                                                                                        |
| <b>CAUSE</b>       | Service hung during initialization, slow disk, network dependency, or inadequate timeout.                                                                                                                                                                                                  |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Increase timeout in unit file: TimeoutStartSec=120</li> <li>• Check why service hangs: strace -p &lt;PID&gt;</li> <li>• Review service logs: journalctl -u &lt;service&gt;</li> <li>• Ensure network dependencies are met before start</li> </ul> |

**#044 systemd: Dependency failed**

|                    |                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A service failed to start because one of its required dependencies could not start.                                                                                                                                                                                                       |
| <b>ROOT CAUSE</b>  | systemd's dependency graph propagates failures; if a Requires= unit fails, dependents are also failed.                                                                                                                                                                                    |
| <b>CAUSE</b>       | A dependent service (database, network, mount) failed to start.                                                                                                                                                                                                                           |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Check: systemctl list-dependencies --failed &lt;service&gt;</li> <li>• Fix the failing dependency first</li> <li>• Use After= instead of Requires= for optional deps</li> <li>• Check journalctl -u &lt;dependency&gt; for root cause</li> </ul> |

**#045 cgroup: Resource limit exceeded**

|                    |                                                                                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A process in a cgroup was killed or throttled because it exceeded the cgroup resource limit.                                                                                                                                                                          |
| <b>ROOT CAUSE</b>  | cgroup v1/v2 enforces memory, CPU, and I/O limits; violations result in OOM kill or throttling.                                                                                                                                                                       |
| <b>CAUSE</b>       | MemoryMax or MemoryLimit in systemd unit too restrictive for workload.                                                                                                                                                                                                |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Check limit: systemctl show &lt;service&gt;   grep Memory</li> <li>• Adjust in unit file: MemoryMax=2G</li> <li>• Override without editing unit: systemctl edit &lt;service&gt;</li> <li>• Monitor: systemd-cgtop</li> </ul> |

**#046 Service: ExecStart not found**

|                    |                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | systemd cannot start a service because the specified executable does not exist.                                                                                                                |
| <b>ROOT CAUSE</b>  | The ExecStart= path in the unit file does not resolve to an existing executable.                                                                                                               |
| <b>CAUSE</b>       | Software not installed, wrong path in unit file, or binary removed.                                                                                                                            |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Check path: which &lt;binary&gt; or ls -la &lt;path&gt;</li> <li>• Install missing package</li> <li>• Correct ExecStart= path in unit file</li> </ul> |

- Run systemctl daemon-reload after editing unit

**#047 journald: No space left for journal**

|             |                                                                                                                                                                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | systemd-journald cannot write new log entries because the journal disk quota is reached.                                                                                                                                                                                 |
| ROOT CAUSE  | journald enforces SystemMaxUse and SystemKeepFree limits on journal storage.                                                                                                                                                                                             |
| CAUSE       | Excessive logging, long retention, or misconfigured journal size limits.                                                                                                                                                                                                 |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check journal size: journalctl --disk-usage</li><li>• Vacuum by time: journalctl --vacuum-time=7d</li><li>• Vacuum by size: journalctl --vacuum-size=1G</li><li>• Adjust /etc/systemd/journald.conf: SystemMaxUse=500M</li></ul> |

**#048 Failed to mount tmpfs**

|             |                                                                                                                                                                                                                                            |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A tmpfs (RAM-backed) filesystem could not be mounted, preventing /tmp or similar from initializing.                                                                                                                                        |
| ROOT CAUSE  | The kernel could not allocate the requested size from available memory for the tmpfs mount.                                                                                                                                                |
| CAUSE       | Insufficient free memory, or size= option exceeds available RAM+swap.                                                                                                                                                                      |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check available memory: free -h</li><li>• Reduce tmpfs size: mount -o remount,size=512M /tmp</li><li>• Add swap if memory is insufficient</li><li>• Review /etc/fstab tmpfs size options</li></ul> |

**#049 Reached target shutdown but service still running**

|             |                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | During shutdown, one or more services did not stop cleanly within the timeout.                                                                                                                                                                                          |
| ROOT CAUSE  | systemd sends SIGTERM during shutdown and waits DefaultTimeoutStopSec; unresponsive services get SIGKILL.                                                                                                                                                               |
| CAUSE       | Service not handling SIGTERM, deadlock, or waiting on I/O.                                                                                                                                                                                                              |
| SOLUTION    | <ul style="list-style-type: none"><li>• Identify stuck service: journalctl -b -1   grep "Stopping Timeout"</li><li>• Fix SIGTERM handling in application</li><li>• Reduce TimeoutStopSec in unit file</li><li>• Pre-stop hook to gracefully drain connections</li></ul> |

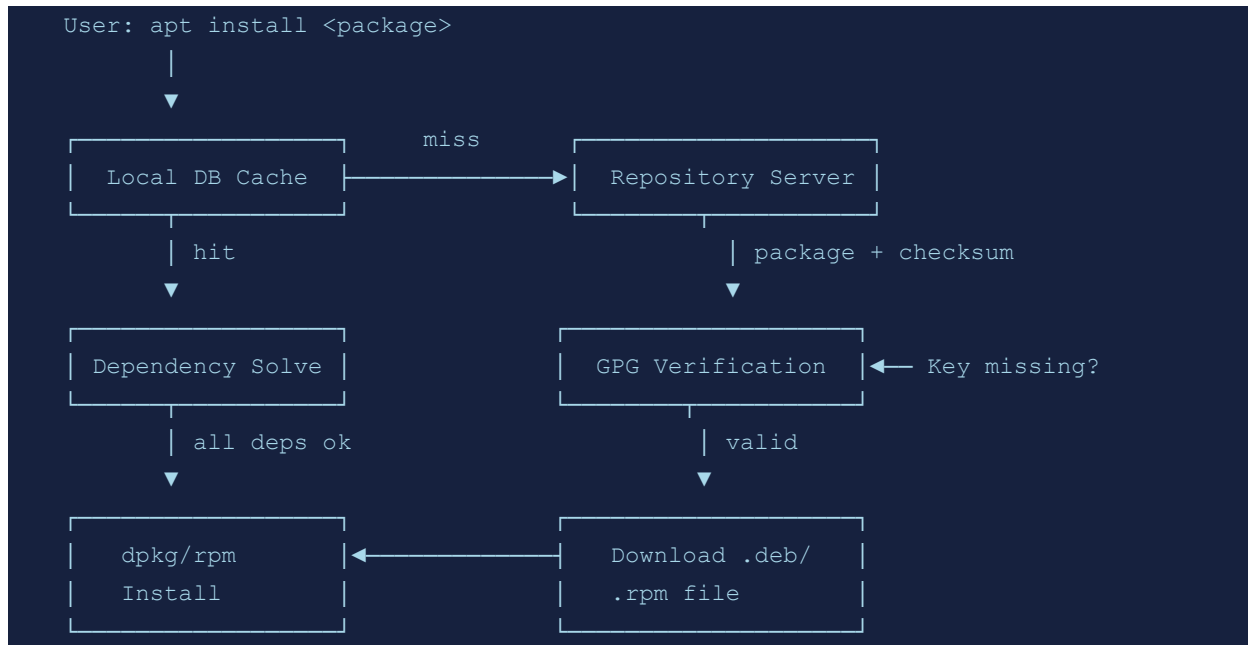
**#050 Timer unit missed activation**

|             |                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A systemd timer unit could not fire at its scheduled time.                                                  |
| ROOT CAUSE  | If the system was off or the timer service failed when the scheduled time passed, the activation is missed. |

|          |                                                                                                                                                                                                                                                                              |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAUSE    | System was powered off, or timer service was disabled.                                                                                                                                                                                                                       |
| SOLUTION | <ul style="list-style-type: none"><li>• Enable persistent timers: Persistent=true in timer unit</li><li>• Check timer status: systemctl list-timers</li><li>• Verify: systemctl is-enabled &lt;timer&gt;.timer</li><li>• Review: journalctl -u &lt;timer&gt;.timer</li></ul> |

## Section 6: Package Management Errors (51–60)

Diagram: Package Manager Resolution Flow (apt/dnf)



### #051 dpkg: dependency problems prevent configuration

|             |                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A Debian package cannot be configured because its dependencies are not satisfied.                                                                                                                                                              |
| ROOT CAUSE  | dpkg's post-installation configuration scripts require dependencies to be fully installed first.                                                                                                                                               |
| CAUSE       | Interrupted install, manually removed dependency, or version conflict.                                                                                                                                                                         |
| SOLUTION    | <ul style="list-style-type: none"><li>Fix with apt: apt-get -f install</li><li>Check broken packages: dpkg --audit</li><li>Force configure: dpkg --configure -a</li><li>Reinstall problematic package: apt reinstall &lt;package&gt;</li></ul> |

### #052 rpm: failed dependencies

|             |                                                                                                                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | RPM refuses to install a package because required dependencies are not present.                                                                                                                                                                                                                                      |
| ROOT CAUSE  | RPM's dependency resolver requires all Requires: entries to be satisfied before installation.                                                                                                                                                                                                                        |
| CAUSE       | Missing dependency, wrong architecture, or repository not enabled.                                                                                                                                                                                                                                                   |
| SOLUTION    | <ul style="list-style-type: none"><li>Use dnf which auto-resolves: dnf install &lt;package&gt;</li><li>Enable required repo: dnf config-manager --enable &lt;repo&gt;</li><li>Force without deps (risky): rpm --nodeps -i &lt;package&gt;.rpm</li><li>Check: rpm -qpR &lt;package&gt;.rpm for requirements</li></ul> |



**#053 apt: Unable to lock the administration directory**

|             |                                                                                                                                                                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The apt/dpkg lock file is held by another process, preventing concurrent package operations.                                                                                                                                                                                                                    |
| ROOT CAUSE  | dpkg uses /var/lib/dpkg/lock-frontent to prevent concurrent modifications to the package database.                                                                                                                                                                                                              |
| CAUSE       | Another apt/dpkg process running (including unattended-upgrades), or stale lock.                                                                                                                                                                                                                                |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check for running process: <code>ps aux   grep -E "apt dpkg"</code></li><li>• Wait for completion, or kill if stale</li><li>• Remove stale lock (only if no apt running): <code>rm /var/lib/dpkg/lock*</code></li><li>• Run: <code>dpkg --configure -a</code></li></ul> |

**#054 yum/dnf: No package available**

|             |                                                                                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The package manager cannot find the requested package in any enabled repository.                                                                                                                                                                                                                        |
| ROOT CAUSE  | No repository in the enabled set has metadata for the requested package name.                                                                                                                                                                                                                           |
| CAUSE       | Typo in package name, package in disabled repo, or EPEL/third-party repo needed.                                                                                                                                                                                                                        |
| SOLUTION    | <ul style="list-style-type: none"><li>• Search: <code>dnf search &lt;keyword&gt;</code></li><li>• Enable EPEL: <code>dnf install epel-release</code></li><li>• Check all repos: <code>dnf repolist all</code></li><li>• Add third-party repo if needed</li><li>• Verify package name and arch</li></ul> |

**#055 pip: Could not find a version that satisfies requirement**

|             |                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | pip cannot find a package version matching the specified requirement in PyPI or configured indexes.                                                                                                                                                                                                                                                                             |
| ROOT CAUSE  | pip queries configured indexes and finds no release matching the version specifier and Python version.                                                                                                                                                                                                                                                                          |
| CAUSE       | Package name typo, network issue, Python version incompatibility, or private PyPI not configured.                                                                                                                                                                                                                                                                               |
| SOLUTION    | <ul style="list-style-type: none"><li>• Search: <code>pip search &lt;package&gt;</code> or check <a href="https://pypi.org">pypi.org</a></li><li>• Specify compatible Python: <code>pip3 install</code> or <code>python3.9 -m pip</code></li><li>• Add index: <code>pip install --index-url &lt;url&gt; &lt;package&gt;</code></li><li>• Check network/proxy settings</li></ul> |

**#056 GPG key verification failed**

|             |                                                                                         |
|-------------|-----------------------------------------------------------------------------------------|
| DESCRIPTION | Package manager refuses to install a package because GPG signature verification failed. |
|-------------|-----------------------------------------------------------------------------------------|

|                   |                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ROOT CAUSE</b> | The package's cryptographic signature does not match the signing key in the keyring, indicating possible tampering.                                                                                                                                                                                                                                                                      |
| <b>CAUSE</b>      | Key not imported, expired key, or package from unofficial source.                                                                                                                                                                                                                                                                                                                        |
| <b>SOLUTION</b>   | <ul style="list-style-type: none"> <li>• Import key: <code>apt-key add &lt;keyfile&gt;</code> or <code>rpm --import &lt;keyfile&gt;</code></li> <li>• For apt: <code>curl -fsSL &lt;key_url&gt;   gpg --dearmor   tee /etc/apt/keyrings/&lt;name&gt;.gpg</code></li> <li>• Verify key fingerprint from official source</li> <li>• Check key expiry: <code>apt-key list</code></li> </ul> |

**#057 Repository metadata outdated**

|                    |                                                                                                                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | Package manager reports that cached repository metadata is stale or corrupt.                                                                                                                                                                                                                                 |
| <b>ROOT CAUSE</b>  | The local metadata cache does not match what the repository server has, causing inconsistency.                                                                                                                                                                                                               |
| <b>CAUSE</b>       | Network failure during metadata sync, or repository was updated since last sync.                                                                                                                                                                                                                             |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Update cache: <code>apt update</code> or <code>dnf makecache</code></li> <li>• Clean cache first: <code>dnf clean all &amp;&amp; dnf makecache</code></li> <li>• Check repo URLs in <code>/etc/yum.repos.d/</code> or <code>/etc/apt/sources.list</code></li> </ul> |

**#058 snap: snap is already installed**

|                    |                                                                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A snap installation fails because the snap is already present in the system.                                                                                                                                                                   |
| <b>ROOT CAUSE</b>  | snapped tracks installed snaps and rejects duplicate installs.                                                                                                                                                                                 |
| <b>CAUSE</b>       | Attempting to install an already installed snap without using refresh.                                                                                                                                                                         |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Update instead: <code>snap refresh &lt;snapname&gt;</code></li> <li>• Check installed: <code>snap list</code></li> <li>• Remove first if needed: <code>snap remove &lt;snapname&gt;</code></li> </ul> |

**#059 pip: externally managed environment**

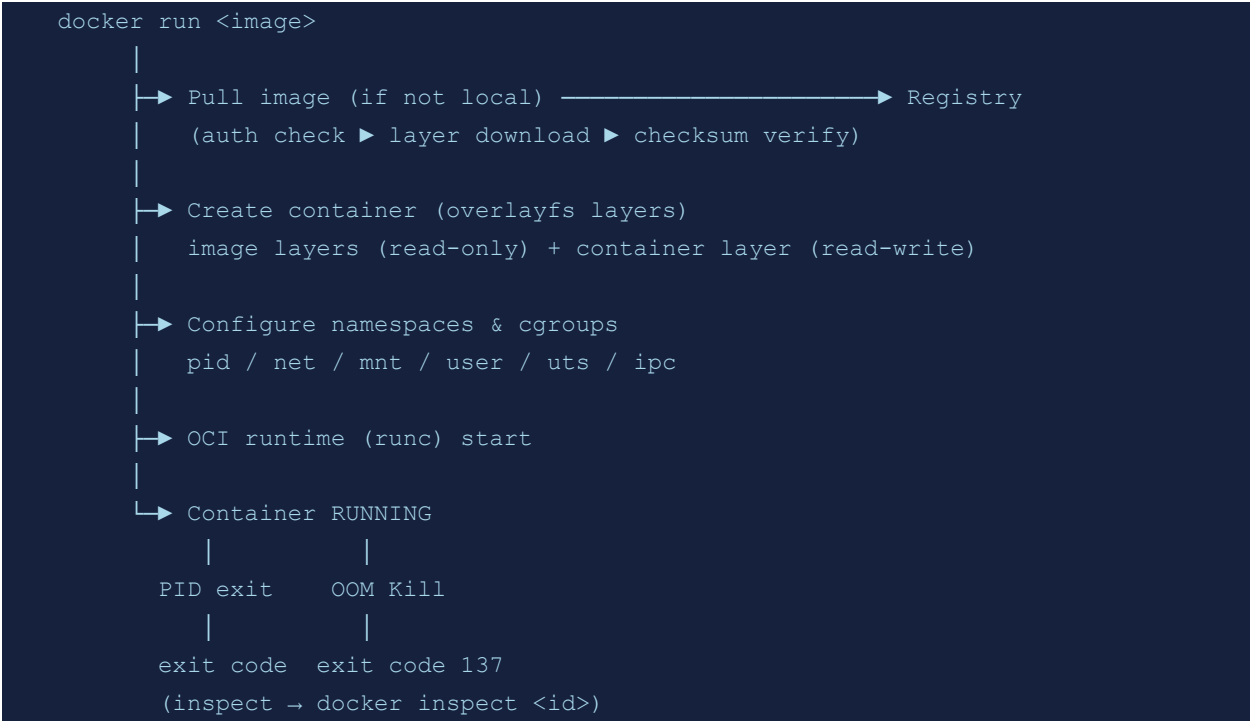
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | pip refuses to install packages system-wide because the Python installation is managed by the OS package manager.                                                                                                                                                                                                                                                                                                                                  |
| <b>ROOT CAUSE</b>  | PEP 668 requires pip to respect OS package manager ownership to prevent conflicts.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>CAUSE</b>       | Trying to pip install on Python managed by apt/dnf without a virtual environment.                                                                                                                                                                                                                                                                                                                                                                  |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Use virtual environment: <code>python3 -m venv .venv &amp;&amp; source .venv/bin/activate</code></li> <li>• Or use <code>--user</code> flag: <code>pip install --user &lt;package&gt;</code></li> <li>• Or use <code>pipx</code> for CLI tools: <code>pipx install &lt;tool&gt;</code></li> <li>• Override (not recommended): <code>pip install --break-system-packages &lt;package&gt;</code></li> </ul> |

**#060 checksum mismatch during download**

|                    |                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | The downloaded package file's checksum does not match the expected value from the repository metadata.                                                                                                                                                                                                                                                              |
| <b>ROOT CAUSE</b>  | The package manager verifies SHA256 checksums against repository metadata; a mismatch indicates corruption.                                                                                                                                                                                                                                                         |
| <b>CAUSE</b>       | Network corruption, CDN cache issue, or man-in-the-middle attack.                                                                                                                                                                                                                                                                                                   |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Re-download: <code>dnf clean packages &amp;&amp; dnf install &lt;package&gt;</code></li><li>• Try different mirror: <code>dnf --disablerepo=* --enablerepo=&lt;specific&gt; install &lt;package&gt;</code></li><li>• Check network for packet corruption</li><li>• Verify no MITM: compare with official checksum</li></ul> |

Section 7: Docker & Container Errors (61–70)

Diagram: Docker Container Lifecycle



| #061 docker: Cannot connect to the Docker daemon |                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                                      | The Docker client cannot communicate with the Docker daemon socket.                                                                                                                                                                                                                                            |
| ROOT CAUSE                                       | The Docker client connects to /var/run/docker.sock; if the daemon is stopped or the socket has wrong permissions, connection fails.                                                                                                                                                                            |
| CAUSE                                            | Docker daemon not running, user not in docker group, or socket permission issue.                                                                                                                                                                                                                               |
| SOLUTION                                         | <ul style="list-style-type: none"><li>• Check daemon: systemctl status docker</li><li>• Start daemon: systemctl start docker</li><li>• Add user to docker group: usermod -aG docker \$USER</li><li>• Log out/in for group change to take effect</li><li>• Verify socket: ls -la /var/run/docker.sock</li></ul> |

| #062 docker: No space left on device |                                                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                          | Docker operations fail because the Docker storage backend has run out of disk space.                                                   |
| ROOT CAUSE                           | Docker stores images, containers, and volumes in /var/lib/docker; this partition can fill up with unused resources.                    |
| CAUSE                                | Accumulated unused images, stopped containers, or large volumes.                                                                       |
| SOLUTION                             | <ul style="list-style-type: none"><li>• Check: docker system df</li><li>• Prune all unused: docker system prune -a --volumes</li></ul> |

- Remove images: `docker image prune -a`
- Remove containers: `docker container prune`
- Expand `/var/lib/docker` partition or move to larger disk

**#063 OCI runtime: permission denied**

|                    |                                                                                                                                                                                                                                                                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | Docker or containerd cannot start a container due to a permission error in the OCI runtime.                                                                                                                                                                                                                                                            |
| <b>ROOT CAUSE</b>  | The container runtime (runc) encounters a permission denial when setting up namespaces, cgroups, or filesystem mounts.                                                                                                                                                                                                                                 |
| <b>CAUSE</b>       | SELinux/AppArmor policy, wrong user namespace configuration, or missing kernel capabilities.                                                                                                                                                                                                                                                           |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Check: <code>docker logs &lt;container&gt;</code></li> <li>• Inspect SELinux: <code>ausearch -m AVC   tail</code></li> <li>• Add SELinux label: <code>docker run --security-opt label=disable</code></li> <li>• Check if user namespaces are enabled: <code>sysctl user.max_user_namespaces</code></li> </ul> |

**#064 docker build: COPY failed**

|                    |                                                                                                                                                                                                                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A Dockerfile COPY instruction fails during build.                                                                                                                                                                                                                                                                          |
| <b>ROOT CAUSE</b>  | The build context does not contain the specified source path, or the destination path cannot be written.                                                                                                                                                                                                                   |
| <b>CAUSE</b>       | <code>.dockerignore</code> excluding needed files, wrong source path, or insufficient disk in build layer.                                                                                                                                                                                                                 |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Check <code>.dockerignore</code> is not excluding needed files</li> <li>• Verify source path relative to Dockerfile location</li> <li>• Use <code>--no-cache</code> to debug: <code>docker build --no-cache</code></li> <li>• Check available disk space on build host</li> </ul> |

**#065 Container exits with code 137**

|                    |                                                                                                                                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A Docker container exits immediately with code 137, indicating it was killed by SIGKILL.                                                                                                                                                                                                                              |
| <b>ROOT CAUSE</b>  | Exit code 137 = 128 + 9 (SIGKILL); the container's main process received SIGKILL from the OOM killer or Docker.                                                                                                                                                                                                       |
| <b>CAUSE</b>       | Container exceeded its memory limit, OOM kill, or docker stop timing out.                                                                                                                                                                                                                                             |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Check: <code>docker inspect &lt;container&gt;   grep OOMKilled</code></li> <li>• Increase memory limit: <code>docker run -m 2g</code></li> <li>• Check logs before exit: <code>docker logs &lt;container&gt;</code></li> <li>• Profile memory usage and fix leaks</li> </ul> |

**#066 docker pull: unauthorized authentication required**

|             |                                                                                                                                                                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A docker pull fails with an authorization error from the registry.                                                                                                                                                                                                                                               |
| ROOT CAUSE  | The registry returned HTTP 401, indicating the client is not authenticated or credentials are invalid.                                                                                                                                                                                                           |
| CAUSE       | Not logged in to registry, expired credentials, or wrong registry URL.                                                                                                                                                                                                                                           |
| SOLUTION    | <ul style="list-style-type: none"><li>• Login: <code>docker login &lt;registry&gt;</code></li><li>• Check credentials: <code>cat ~/.docker/config.json</code></li><li>• For ECR: <code>aws ecr get-login-password   docker login</code></li><li>• Verify image name includes correct registry hostname</li></ul> |

**#067 Kubernetes: CrashLoopBackOff**

|             |                                                                                                                                                                                                                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A Kubernetes pod repeatedly crashes and is restarted by kubelet in an exponential backoff loop.                                                                                                                                                                                                                                                       |
| ROOT CAUSE  | The container's main process exits with a non-zero code, causing kubelet to restart it with increasing delays.                                                                                                                                                                                                                                        |
| CAUSE       | Application error, missing environment variable, failed dependency, or OOM kill.                                                                                                                                                                                                                                                                      |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check logs: <code>kubectl logs &lt;pod&gt; --previous</code></li><li>• Describe pod: <code>kubectl describe pod &lt;pod&gt;</code></li><li>• Check resource limits: <code>kubectl get pod &lt;pod&gt; -o yaml   grep -A5 resources</code></li><li>• Fix application crash or adjust resource limits</li></ul> |

**#068 Kubernetes: ImagePullBackOff**

|             |                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | Kubernetes cannot pull the container image for a pod.                                                                                                                                                                                                                                                                                                                                        |
| ROOT CAUSE  | kubelet failed to pull the image and is in exponential backoff, retrying after increasing delays.                                                                                                                                                                                                                                                                                            |
| CAUSE       | Wrong image name/tag, private registry without credentials, or registry unreachable.                                                                                                                                                                                                                                                                                                         |
| SOLUTION    | <ul style="list-style-type: none"><li>• Describe pod: <code>kubectl describe pod &lt;pod&gt;</code></li><li>• Create registry secret: <code>kubectl create secret docker-registry</code></li><li>• Reference in pod: <code>imagePullSecrets</code></li><li>• Verify image exists: <code>docker pull &lt;image&gt; from node</code></li><li>• Check registry DNS from cluster nodes</li></ul> |

**#069 docker network: endpoint with name already exists**

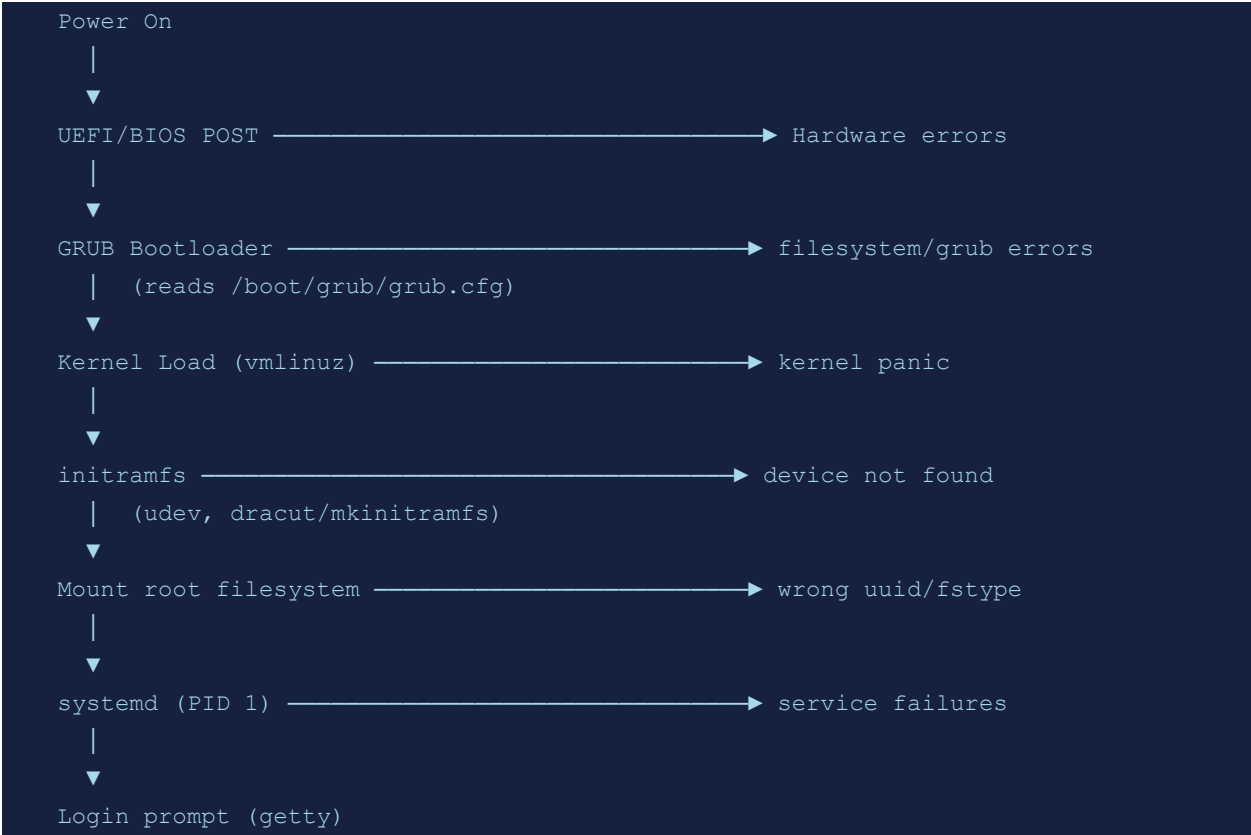
|             |                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------|
| DESCRIPTION | Docker network creation or container connection fails because the endpoint name is already registered. |
| ROOT CAUSE  | Docker's network model requires unique endpoint names within a network namespace.                      |
| CAUSE       | Leftover endpoint from previous container that was not properly removed.                               |

|                 |                                                                                                                                                                                                                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SOLUTION</b> | <ul style="list-style-type: none"><li>• Disconnect stale endpoint: docker network disconnect -f &lt;network&gt; &lt;container&gt;</li><li>• List endpoints: docker network inspect &lt;network&gt;</li><li>• Prune networks: docker network prune</li><li>• Recreate network if needed</li></ul> |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| #070 containerd: failed to create shim |                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b>                     | containerd cannot start the container shim process, preventing container startup.                                                                                                                                                                                           |
| <b>ROOT CAUSE</b>                      | The containerd-shim binary cannot be executed due to permission, missing binary, or cgroup issues.                                                                                                                                                                          |
| <b>CAUSE</b>                           | Missing shim binary, incompatible versions between containerd and runc, or SELinux denial.                                                                                                                                                                                  |
| <b>SOLUTION</b>                        | <ul style="list-style-type: none"><li>• Check containerd logs: journalctl -u containerd</li><li>• Verify shim binary: which containerd-shim-runc-v2</li><li>• Check cgroup version compatibility</li><li>• Reinstall containerd if binary is missing or corrupted</li></ul> |

Section 8: Kernel & Boot Errors (71–80)

Diagram: Linux Boot Sequence & Error Points



| #071 Kernel panic: not syncing |                                                                                                                                                                                                                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                    | The kernel encountered an unrecoverable error and halted the system.                                                                                                                                                                                                                      |
| ROOT CAUSE                     | A kernel panic occurs when the kernel cannot continue safely. It dumps state and stops.                                                                                                                                                                                                   |
| CAUSE                          | Corrupted kernel, missing initramfs, hardware failure, or critical driver bug.                                                                                                                                                                                                            |
| SOLUTION                       | <ul style="list-style-type: none"><li>• Boot from rescue disk</li><li>• Check boot logs for panic message</li><li>• Reinstall kernel: dnf reinstall kernel</li><li>• Rebuild initramfs: dracut --force or update-initramfs -u</li><li>• Check hardware for failures: memtest86+</li></ul> |

| #072 GRUB: error: unknown filesystem |                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                          | GRUB fails to read the boot partition and cannot load the kernel.                                                        |
| ROOT CAUSE                           | GRUB modules attempt to read the filesystem where /boot resides; an unrecognized or corrupted filesystem causes failure. |
| CAUSE                                | Filesystem corruption, GRUB modules missing, or wrong root device specified.                                             |



**SOLUTION**

- Boot GRUB rescue mode
- Set root: set root=(hd0,1)
- Load modules: insmod ext2
- Boot: linux /boot/vmlinuz root=/dev/sda1; initrd /boot/initrd.img; boot
- After boot: grub-install /dev/sda && update-grub

**#073 initramfs: Unable to find a medium containing a live file system****DESCRIPTION**

The initramfs cannot find and mount the root filesystem during early boot.

**ROOT CAUSE**

The initial ramdisk cannot resolve the root= parameter to an accessible block device.

**CAUSE**

Wrong root= in GRUB, UUID mismatch, missing storage driver in initramfs.

**SOLUTION**

- Verify UUID: blkid | grep <root\_partition>
- Compare with /etc/fstab UUID
- Update /etc/default/grub with correct root
- Rebuild initramfs with correct modules

**#074 modprobe: Module not found****DESCRIPTION**

The kernel module loader cannot find the specified module in the module search paths.

**ROOT CAUSE**

modprobe searches /lib/modules/\$(uname -r)/ for the module; if absent, it fails.

**CAUSE**

Module not installed for current kernel, misspelled module name, or kernel upgrade without matching modules.

**SOLUTION**

- Verify kernel version: uname -r
- Check module: find /lib/modules/\$(uname -r) -name "<module>.\*"
- Install kernel modules: apt install linux-modules-extra-\$(uname -r)
- Run: depmod -a to rebuild module dependency db

**#075 kernel: NMI watchdog: BUG: soft lockup****DESCRIPTION**

The NMI watchdog detected a CPU stuck in kernel code for more than 20 seconds.

**ROOT CAUSE**

The NMI watchdog fires a hardware interrupt every 20s; if a CPU does not respond (held by spinlock or IRQ disabled), a soft lockup BUG is recorded.

**CAUSE**

Kernel bug, driver hanging with IRQs disabled, or extreme CPU saturation.

**SOLUTION**

- Check dmesg for lockup trace
- Identify module from stack trace
- Update kernel and affected driver
- File upstream kernel bug report with trace

**#076 ACPI: Warning: Unsupported BusWidth**

|             |                                                                                                                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The kernel ACPI subsystem logs warnings about unsupported bus width in ACPI tables.                                                                                                                                                                            |
| ROOT CAUSE  | The ACPI firmware tables define hardware resources; inconsistencies trigger kernel warnings.                                                                                                                                                                   |
| CAUSE       | BIOS/UEFI firmware bug, or running newer kernel on hardware with outdated firmware.                                                                                                                                                                            |
| SOLUTION    | <ul style="list-style-type: none"><li>• Update BIOS/UEFI firmware</li><li>• Add kernel parameter to suppress: <code>acpi=off</code> (test only)</li><li>• Check if upstream kernel has fix for your hardware</li><li>• File bug with firmware vendor</li></ul> |

**#077 dracut: Cannot find device (root)**

|             |                                                                                                                                                                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The dracut-based initramfs cannot locate the root block device during boot.                                                                                                                                                                                                                                           |
| ROOT CAUSE  | dracut's udev rules and device discovery cannot find the expected root device by UUID, label, or path.                                                                                                                                                                                                                |
| CAUSE       | UUID changed after disk replacement, missing storage driver, or corrupted udev rules.                                                                                                                                                                                                                                 |
| SOLUTION    | <ul style="list-style-type: none"><li>• Boot to dracut emergency shell</li><li>• Manually mount root: <code>mount /dev/sda1 /sysroot</code></li><li>• Check <code>/etc/fstab</code> for correct UUID</li><li>• Rebuild initramfs: <code>dracut --force /boot/initramfs-\$(uname -r).img \$(uname -r)</code></li></ul> |

**#078 EFI: boot entry not found**

|             |                                                                                                                                                                                                                                                                                                                                            |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The UEFI firmware cannot find the configured boot entry and falls back to removable media or fails.                                                                                                                                                                                                                                        |
| ROOT CAUSE  | UEFI boot entries stored in NVRAM reference EFI applications on the ESP partition; missing/corrupt entries cause boot failure.                                                                                                                                                                                                             |
| CAUSE       | Boot entry deleted, NVRAM reset, ESP corruption, or bootloader removed.                                                                                                                                                                                                                                                                    |
| SOLUTION    | <ul style="list-style-type: none"><li>• Boot from USB recovery</li><li>• List entries: <code>efibootmgr -v</code></li><li>• Reinstall bootloader: <code>grub-install --target=x86_64-efi</code></li><li>• Create entry: <code>efibootmgr --create --disk /dev/sda --part 1 --label Linux --loader /EFI/ubuntu/grubx64.efi</code></li></ul> |

**#079 kernel: SCSI error: return code = 0x08000002**

|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| DESCRIPTION | The kernel SCSI subsystem reports a device error for a storage device command. |
|-------------|--------------------------------------------------------------------------------|

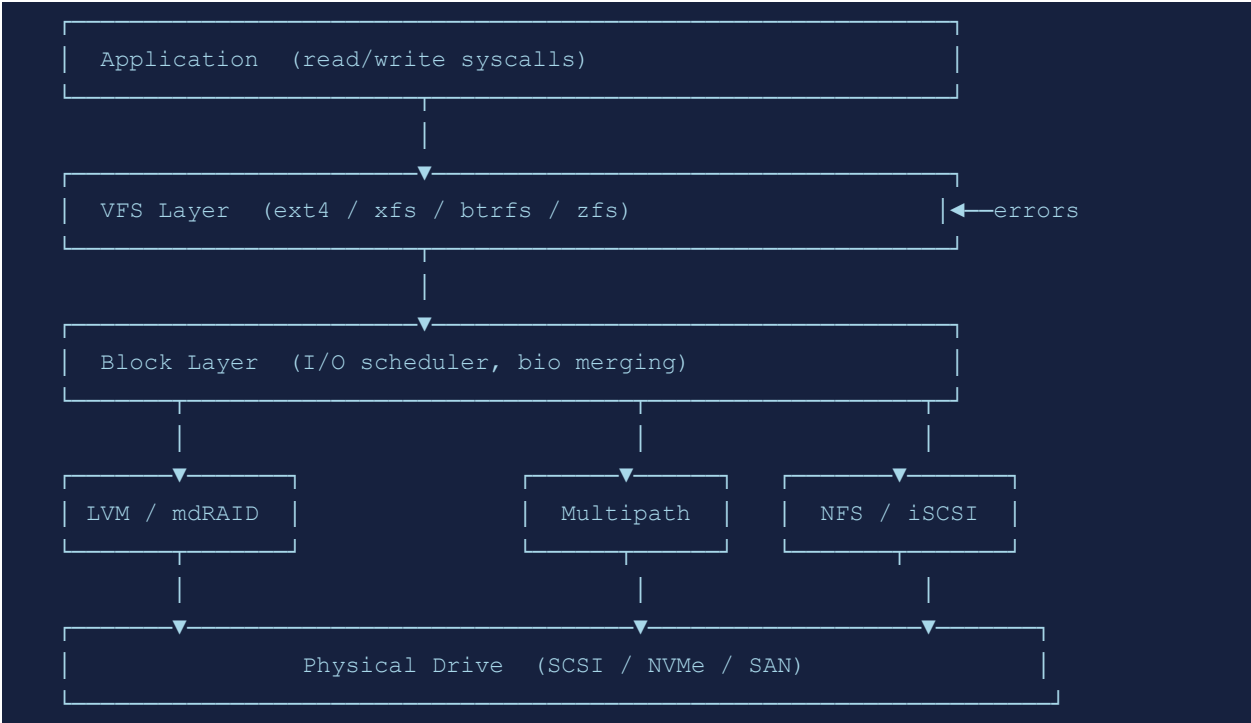
|                   |                                                                                                                                                                                                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ROOT CAUSE</b> | SCSI mid-layer received an error from the HBA or device; 0x08000002 often indicates DID_TRANSPORT_DISRUPTED.                                                                                                                                                                                      |
| <b>CAUSE</b>      | Failing HBA, loose SAS/SATA cable, enclosure issue, or SAN path failure.                                                                                                                                                                                                                          |
| <b>SOLUTION</b>   | <ul style="list-style-type: none"><li>• Check: <code>dmesg   grep -i scsi</code></li><li>• Run storage diagnostics: <code>smartctl -a /dev/sdX</code></li><li>• Check SAN/storage array event logs</li><li>• Reseat cables and HBA</li><li>• Enable multipath if not already configured</li></ul> |

#### #080 memory: EDAC error detected

|                    |                                                                                                                                                                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | The EDAC (Error Detection and Correction) subsystem reports correctable or uncorrectable memory errors.                                                                                                                                                                                                  |
| <b>ROOT CAUSE</b>  | EDAC hardware monitors ECC RAM for bit errors; single-bit errors are corrected, multi-bit are uncorrectable and can panic.                                                                                                                                                                               |
| <b>CAUSE</b>       | Failing RAM DIMM, poor seating, or power supply instability.                                                                                                                                                                                                                                             |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Check: <code>edac-util -s 4</code> or <code>/sys/bus/platform/drivers/i7core_edac/</code></li><li>• Run <code>memtest86+</code> to identify failing DIMM slot</li><li>• Replace failing DIMM</li><li>• Check kernel: <code>dmesg   grep -i edac</code></li></ul> |

Section 9: Storage & RAID Errors (81–90)

Diagram: Storage Stack: From Application to Disk



| #081 LVM: Couldn't find device with uuid |                                                                                                                                                                                                                                                                 |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                              | LVM cannot find a Physical Volume by its stored UUID, preventing Volume Group activation.                                                                                                                                                                       |
| ROOT CAUSE                               | LVM scans configured devices for PV labels; if the device is missing or renamed, the VG cannot be fully assembled.                                                                                                                                              |
| CAUSE                                    | Disk removed, device renamed after udev rule change, or disk failure.                                                                                                                                                                                           |
| SOLUTION                                 | <ul style="list-style-type: none"><li>• Scan for PVs: pvs and pvdisplay</li><li>• Check device availability: lsblk</li><li>• Update device path in LVM config if renamed</li><li>• Remove missing PV (data loss): vgreduce --removemissing &lt;vg&gt;</li></ul> |

| #082 mdadm: ARRAY failed to assemble |                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                          | A software RAID array cannot be assembled because required member disks are missing or degraded.                            |
| ROOT CAUSE                           | mdadm requires a minimum number of disks to assemble an array; below that, it refuses to assemble to avoid data corruption. |
| CAUSE                                | Failed disk, wrong device node, or UUID mismatch in mdadm.conf.                                                             |
| SOLUTION                             | <ul style="list-style-type: none"><li>• Scan arrays: mdadm --examine --scan</li></ul>                                       |

- Force assemble (degraded): `mdadm --assemble --force /dev/md0 /dev/sd[bcde]`
- Check array status: `cat /proc/mdstat`
- Replace failed disk and add: `mdadm /dev/md0 --add /dev/sdX`

**#083 XFS: metadata I/O error**

|                    |                                                                                                                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | XFS logged a metadata I/O error, indicating it cannot read or write filesystem structures.                                                                                                                                                                                                                |
| <b>ROOT CAUSE</b>  | XFS encountered a read or write error on a metadata block, which can lead to filesystem shutdown.                                                                                                                                                                                                         |
| <b>CAUSE</b>       | Failing disk, storage controller error, or bad block.                                                                                                                                                                                                                                                     |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Check dmesg for underlying I/O errors</li> <li>• Run SMART check: <code>smartctl -a /dev/sdX</code></li> <li>• Repair (unmounted): <code>xfs_repair /dev/sdXN</code></li> <li>• Mount with <code>nobarrier</code> as temp workaround on good hardware</li> </ul> |

**#084 ext4: journal commit I/O error**

|                    |                                                                                                                                                                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | The ext4 filesystem's journal cannot commit a transaction due to an I/O error.                                                                                                                                                                                                                                                 |
| <b>ROOT CAUSE</b>  | ext4 uses a journal (journal area on disk) for crash consistency; I/O errors writing to the journal cause filesystem to enter read-only or error state.                                                                                                                                                                        |
| <b>CAUSE</b>       | Failing disk, bad sectors in journal area, or storage timeout.                                                                                                                                                                                                                                                                 |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Check dmesg: <code>dmesg   grep -i ext4</code></li> <li>• Check disk health: <code>smartctl -a /dev/sdX</code></li> <li>• Force fsck: <code>fsck -f /dev/sdXN</code></li> <li>• Disable journaling temporarily (test only): <code>tune2fs -O ^has_journal /dev/sdXN</code></li> </ul> |

**#085 NFS: stale file handle**

|                    |                                                                                                                                                                                                                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | An operation on an NFS-mounted file returns a stale file handle error.                                                                                                                                                                                                                                        |
| <b>ROOT CAUSE</b>  | The file handle stored by the NFS client refers to an inode that no longer exists on the server (deleted or server reboot).                                                                                                                                                                                   |
| <b>CAUSE</b>       | Server rebooted, export changed, or file deleted on server while client held it open.                                                                                                                                                                                                                         |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Unmount and remount NFS share</li> <li>• Remount: <code>umount -f /mnt/nfs &amp;&amp; mount /mnt/nfs</code></li> <li>• Check NFS server: <code>showmount -e &lt;server&gt;</code></li> <li>• Use <code>soft,intr</code> mount options for better recovery</li> </ul> |

**#086 ZFS: pool is faulted**

|             |                                                                                                                                                                                                                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A ZFS storage pool entered the FAULTED state and is offline.                                                                                                                                                                                                                                                                            |
| ROOT CAUSE  | ZFS fault management marks a pool faulted when too many vdevs are unavailable to maintain the redundancy level.                                                                                                                                                                                                                         |
| CAUSE       | Multiple disk failures exceeding RAID-Z redundancy, or VDEV corruption.                                                                                                                                                                                                                                                                 |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check: zpool status</li><li>• Clear errors if hardware replaced: zpool clear &lt;pool&gt;</li><li>• Replace disk: zpool replace &lt;pool&gt; &lt;old&gt; &lt;new&gt;</li><li>• Import from cache: zpool import -f &lt;pool&gt;</li><li>• Restore from backup if pool is unrecoverable</li></ul> |

### #087 LUKS: Failed to open the device

|             |                                                                                                                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A LUKS-encrypted device cannot be unlocked and opened.                                                                                                                                                                                                                                                                    |
| ROOT CAUSE  | cryptsetup verify the passphrase/keyfile against the LUKS header; failure indicates wrong key or corrupted header.                                                                                                                                                                                                        |
| CAUSE       | Wrong passphrase, corrupted LUKS header, keyfile missing, or slot erased.                                                                                                                                                                                                                                                 |
| SOLUTION    | <ul style="list-style-type: none"><li>• Try all passphrases: cryptsetup open /dev/sdX cryptname</li><li>• Check slots: cryptsetup luksDump /dev/sdX</li><li>• Restore header from backup: cryptsetup luksHeaderRestore</li><li>• Ensure keyfile is present and correct: cryptsetup --key-file=/path/to/key open</li></ul> |

### #088 iSCSI: connection closed

|             |                                                                                                                                                                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | An iSCSI initiator lost its connection to the target, causing storage I/O to fail.                                                                                                                                                                                                                           |
| ROOT CAUSE  | The TCP connection between the initiator and target was interrupted; iSCSI session recovery depends on timers and configuration.                                                                                                                                                                             |
| CAUSE       | Network outage, target service restart, firewall change, or iSCSI target crash.                                                                                                                                                                                                                              |
| SOLUTION    | <ul style="list-style-type: none"><li>• Restart iSCSI initiator: systemctl restart iscsid</li><li>• Reconnect session: iscsiadm -m session --rescan</li><li>• Check network path to iSCSI target</li><li>• Review /var/log/messages for iSCSI errors</li><li>• Tune recovery timers in iscsid.conf</li></ul> |

### #089 multipath: path failed

|             |                                                                                              |
|-------------|----------------------------------------------------------------------------------------------|
| DESCRIPTION | Device Mapper Multipath reports that one or more paths to a storage device have failed.      |
| ROOT CAUSE  | multipathd monitors all paths to a LUN; a path failure triggers failover to surviving paths. |
| CAUSE       | HBA port failure, SAN switch port down, or storage controller path issue.                    |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check: multipathd show paths</li></ul>               |

- Check failed path: multipathd show topology
- Reinstate: multipathd reinstate path <path>
- Fix underlying HBA/SAN issue
- Review /etc/multipath.conf for correct failover settings

### #090 btrfs: parent transid verify failed

|             |                                                                                                                                                                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | btrfs detected a consistency error in its B-tree during a read operation.                                                                                                                                                                                                                              |
| ROOT CAUSE  | btrfs stores generation numbers (transid) to verify tree nodes; a mismatch indicates that an older version of a node was written (write ordering violation).                                                                                                                                           |
| CAUSE       | Disk firmware write caching bug, power loss, or faulty storage controller.                                                                                                                                                                                                                             |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check btrfs: btrfs check /dev/sdXN (read-only first)</li><li>• Repair (may lose data): btrfs check --repair /dev/sdXN</li><li>• Enable metadata redundancy: convert to RAID1 for metadata</li><li>• Replace storage hardware and restore from backup</li></ul> |

Section 10: Security Errors (91–100)

Diagram: Linux Security Layers



| #091 SELinux: avc: denied |                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION               | SELinux blocked an operation because the security policy did not permit it.                                                                                                                                                                                                                                                              |
| ROOT CAUSE                | SELinux's mandatory access control evaluated the source context, target context, and object class against the policy and returned a denial.                                                                                                                                                                                              |
| CAUSE                     | Incorrect file context, new application not covered by policy, or policy not updated after change.                                                                                                                                                                                                                                       |
| SOLUTION                  | <ul style="list-style-type: none"><li>• Check: ausearch -m AVC -ts recent</li><li>• Generate policy module: audit2allow -M mypol &lt; /var/log/audit/audit.log</li><li>• Load module: semodule -i mypol.pp</li><li>• Fix file context: chcon -t &lt;correct_type&gt; /path/to/file</li><li>• Or restore: restorecon -Rv /path/</li></ul> |

| #092 AppArmor: operation not permitted |                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------|
| DESCRIPTION                            | AppArmor denied an operation because no rule in the active profile allows it.                    |
| ROOT CAUSE                             | AppArmor profiles define an allow-list; any operation not listed is denied when in enforce mode. |



|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAUSE    | Application attempting to access path/syscall not in profile, or profile in wrong mode.                                                                                                                                                                                                                                                                                                                                                         |
| SOLUTION | <ul style="list-style-type: none"><li>• Check: <code>dmesg   grep apparmor</code> or <code>/var/log/syslog   grep DENIED</code></li><li>• Set to complain mode: <code>aa-complain /etc/apparmor.d/&lt;profile&gt;</code></li><li>• Generate additions: <code>aa-logprof</code></li><li>• Edit profile: <code>/etc/apparmor.d/&lt;profile&gt;</code></li><li>• Reload: <code>apparmor_parser -r /etc/apparmor.d/&lt;profile&gt;</code></li></ul> |

### #093 Firewallld: Error: ALREADY\_ENABLED

|             |                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | firewall-cmd reports that a rule or service is already enabled in the specified zone.                                                                                                                                                                                                                                                                                                          |
| ROOT CAUSE  | firewalld tracks enabled rules; attempting to add a duplicate returns an error.                                                                                                                                                                                                                                                                                                                |
| CAUSE       | Rule already added, or configuration applied twice by automation.                                                                                                                                                                                                                                                                                                                              |
| SOLUTION    | <ul style="list-style-type: none"><li>• Query current rules: <code>firewall-cmd --list-all</code></li><li>• Check service state: <code>firewall-cmd --query-service=&lt;service&gt;</code></li><li>• Idempotent additions add: <code>--permanent</code> and ignore <code>ALREADY_ENABLED</code></li><li>• Use Ansible <code>firewalld</code> module with <code>state: enabled</code></li></ul> |

### #094 fail2ban: Already banned

|             |                                                                                                                                                                                                                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | fail2ban logs show an IP is already in the ban list but the ban command is attempted again.                                                                                                                                                                                                                                                         |
| ROOT CAUSE  | fail2ban tracks banned IPs in its database; log parsing may trigger a second ban attempt if already present.                                                                                                                                                                                                                                        |
| CAUSE       | High log event rate causing duplicate matches, or database inconsistency.                                                                                                                                                                                                                                                                           |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check ban status: <code>fail2ban-client status &lt;jail&gt;</code></li><li>• Check DB: <code>fail2ban-client get &lt;jail&gt; banip</code></li><li>• Unban if needed: <code>fail2ban-client set &lt;jail&gt; unbanip &lt;ip&gt;</code></li><li>• Review filter regex to prevent duplicate matches</li></ul> |

### #095 openssl: certificate has expired

|             |                                                                                                                                                                                                                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | An SSL/TLS operation fails because the certificate's validity period has ended.                                                                                                                                                                                                                                                |
| ROOT CAUSE  | TLS implementations verify the <code>notAfter</code> field against system clock; past expiry causes handshake failure.                                                                                                                                                                                                         |
| CAUSE       | Certificate not renewed before expiry, or system clock is incorrect.                                                                                                                                                                                                                                                           |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check cert: <code>openssl x509 -noout -dates -in cert.pem</code></li><li>• Check system time: <code>timedatectl</code></li><li>• Renew with Let's Encrypt: <code>certbot renew</code></li><li>• Replace with new certificate</li><li>• Set up monitoring/alerting for expiry</li></ul> |

**#096 SSH: Host key verification failed**

|             |                                                                                                                                                                                                                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The SSH client refuses to connect because the server's host key does not match the stored known_hosts entry.                                                                                                                                                                                              |
| ROOT CAUSE  | SSH clients store server fingerprints; a changed fingerprint could indicate a man-in-the-middle attack.                                                                                                                                                                                                   |
| CAUSE       | Server OS reinstalled (new host key), server IP reused, or actual MITM attack.                                                                                                                                                                                                                            |
| SOLUTION    | <ul style="list-style-type: none"><li>• Verify change is expected (server rebuild)</li><li>• Remove old entry: <code>ssh-keygen -R &lt;hostname&gt;</code></li><li>• Or: <code>sed -i "&lt;hostname&gt;/d" ~/.ssh/known_hosts</code></li><li>• Reconnect and verify new fingerprint out-of-band</li></ul> |

**#097 GPG: No public key**

|             |                                                                                                                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A GPG decryption or verification operation fails because the required public key is not in the keyring.                                                                                                                                                                                                              |
| ROOT CAUSE  | GPG looks up the key ID specified in the encrypted/signed data; if absent from the keyring, it cannot proceed.                                                                                                                                                                                                       |
| CAUSE       | Key not imported, wrong keyring, or key server unreachable.                                                                                                                                                                                                                                                          |
| SOLUTION    | <ul style="list-style-type: none"><li>• Import from keyserver: <code>gpg --keyserver keys.openpgp.org --recv-keys &lt;keyid&gt;</code></li><li>• Import from file: <code>gpg --import public.key</code></li><li>• List keyring: <code>gpg --list-keys</code></li><li>• Verify key fingerprint after import</li></ul> |

**#098 auditd: backlog limit exceeded**

|             |                                                                                                                                                                                                                                                                                    |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The kernel audit subsystem dropped audit events because the backlog queue overflowed.                                                                                                                                                                                              |
| ROOT CAUSE  | auditd has a configurable backlog limit; if the kernel generates events faster than auditd reads them, overflow occurs.                                                                                                                                                            |
| CAUSE       | Very high syscall rate, audit rules too broad, or auditd disk I/O slow.                                                                                                                                                                                                            |
| SOLUTION    | <ul style="list-style-type: none"><li>• Increase backlog: <code>auditctl -b 8192</code></li><li>• Set in <code>/etc/audit/auditd.conf</code>: <code>tcp_listen_queue=100</code></li><li>• Narrow audit rules to reduce event volume</li><li>• Speed up audit log storage</li></ul> |

**#099 rkhunter: Warning: suspicious file found**

|             |                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | rkhunter detected a file that matches its database of known rootkits or suspicious file properties.        |
| ROOT CAUSE  | rkhunter compares file hashes, SUID bits, hidden files, and known rootkit signatures against its database. |

|          |                                                                                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAUSE    | Rootkit infection, false positive from legitimate software change, or outdated rkhunter database.                                                                                                                                                                                                                          |
| SOLUTION | <ul style="list-style-type: none"><li>• Review: rkhunter --report-warnings-only</li><li>• Update database: rkhunter --update</li><li>• Whitelist false positive in /etc/rkhunter.conf</li><li>• If true positive: isolate system, restore from known-good backup</li><li>• Forensic analysis with chkrootkit too</li></ul> |

### #100 chroot: failed to run command

|             |                                                                                                                                                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A chroot operation fails to execute a command inside the chroot environment.                                                                                                                                                                                                                                           |
| ROOT CAUSE  | The chroot changes the root directory; if required libraries, the dynamic linker, or the binary itself are missing inside the chroot, execution fails.                                                                                                                                                                 |
| CAUSE       | Incomplete chroot setup, missing /lib or /lib64, or binary not present inside chroot.                                                                                                                                                                                                                                  |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check binary exists: ls &lt;chroot&gt;/usr/bin/&lt;binary&gt;</li><li>• Copy required libs: ldd &lt;binary&gt; to find deps</li><li>• Mount proc/sys/dev: mount --bind /dev &lt;chroot&gt;/dev</li><li>• Or use schroot/debootstrap for complete chroot environments</li></ul> |

## Section 11: Advanced Networking (101–110)

### Diagram: Advanced Network Troubleshooting Path

```

Is the issue PHYSICAL?
├─ ip link show → state DOWN? → ip link set <if> up
└─ ethtool <if> → Link detected: no → cable/NIC issue
    |
    IP LAYER
    ├─ ip addr show → IP assigned? → dhclient / static config
    └─ ip route show → default route? → ip route add default via <gw>
        |
        TRANSPORT LAYER
        ├─ ping <dst> → ICMP passes? → no → firewall?
        ├─ traceroute → where does it stop?
        └─ ss -tlnp → service listening on correct port?
            |
            APPLICATION LAYER
            ├─ curl -v → SSL cert valid? → DNS resolving?
            └─ tcpdump -i <if> port <n> → packets arriving?
  
```

### #101 iptables: invalid source/destination

|             |                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | An iptables rule addition fails because the IP address or CIDR notation is invalid.                                                                                                                                       |
| ROOT CAUSE  | iptables parses address specifications strictly; incorrect notation causes rule rejection.                                                                                                                                |
| CAUSE       | Typo in IP, incorrect CIDR prefix, or using IPv6 address with iptables (use ip6tables).                                                                                                                                   |
| SOLUTION    | <ul style="list-style-type: none"> <li>• Validate IP: <code>ipcalc &lt;ip&gt;/&lt;prefix&gt;</code></li> <li>• Use ip6tables for IPv6 rules</li> <li>• Check CIDR: ensure prefix length matches address family</li> </ul> |

### #102 tc: Cannot find device

|             |                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The tc (traffic control) command cannot find the specified network interface.                                                                                                                                                                    |
| ROOT CAUSE  | tc operates on kernel network interfaces by name; the specified interface does not exist.                                                                                                                                                        |
| CAUSE       | Interface down, renamed, or not yet created (virtual interface).                                                                                                                                                                                 |
| SOLUTION    | <ul style="list-style-type: none"> <li>• List interfaces: <code>ip link show</code></li> <li>• Check interface state: <code>ip link show &lt;iface&gt;</code></li> <li>• Ensure virtual interface is created first (for tunnels/veth)</li> </ul> |

### #103 nftables: Error: no such file or directory

|             |                                                                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | nftables ruleset loading fails because a referenced file (for sets or include) does not exist.                                                                                                                                         |
| ROOT CAUSE  | nft processes include directives; missing files cause a parse/load error.                                                                                                                                                              |
| CAUSE       | Ruleset file deleted, wrong path in nft config, or first-run without baseline file.                                                                                                                                                    |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check nft file paths: <code>nft -c -f /etc/nftables.conf</code></li><li>• Create missing file with appropriate content</li><li>• List current ruleset: <code>nft list ruleset</code></li></ul> |

#### #104 VPN: TLS handshake failed

|             |                                                                                                                                                                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A VPN client (OpenVPN or similar) fails to establish a TLS connection with the server.                                                                                                                                                                                                                         |
| ROOT CAUSE  | The TLS handshake process failed due to certificate, cipher, or version mismatch.                                                                                                                                                                                                                              |
| CAUSE       | Expired certificate, unsupported cipher suite, TLS version mismatch, or firewall blocking.                                                                                                                                                                                                                     |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check VPN logs: <code>journalctl -u openvpn</code></li><li>• Verify certificate validity: <code>openssl verify client.crt</code></li><li>• Align <code>tls-version-min</code> settings between client and server</li><li>• Ensure UDP/TCP port is accessible</li></ul> |

#### #105 hostname: Name or service not known

|             |                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A hostname cannot be resolved via any configured resolution mechanism.                                                                                                                                                                                                                                                                                                    |
| ROOT CAUSE  | The resolver tried DNS, <code>/etc/hosts</code> , and configured NSS plugins; none returned a result for the hostname.                                                                                                                                                                                                                                                    |
| CAUSE       | Hostname not in DNS, <code>/etc/hosts</code> , or search domain not configured.                                                                                                                                                                                                                                                                                           |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check <code>/etc/hosts</code> for static entries</li><li>• Verify <code>/etc/resolv.conf</code> search domain</li><li>• Test: <code>dig &lt;hostname&gt;</code> vs <code>dig &lt;hostname&gt;.&lt;domain&gt;</code></li><li>• Add to <code>/etc/hosts</code> as temporary fix: <code>&lt;ip&gt; &lt;hostname&gt;</code></li></ul> |

#### #106 socket: Too many open sockets

|             |                                                                                                                                                                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A service cannot create a new socket because the per-process or system socket limit is reached.                                                                                                                                                                                                        |
| ROOT CAUSE  | Sockets consume file descriptors; hitting the fd limit also prevents new socket creation.                                                                                                                                                                                                              |
| CAUSE       | Socket leak in application, too many concurrent connections, or fd limit too low.                                                                                                                                                                                                                      |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check: <code>ss -s</code> for socket summary</li><li>• Increase fd limit: <code>ulimit -n 65536</code></li><li>• Check for socket leaks: <code>lssof -p &lt;PID&gt;   grep socket   wc -l</code></li><li>• Fix application to properly close sockets</li></ul> |

**#107 NetworkManager: device not managed**

|             |                                                                                                                                                                                                                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | NetworkManager reports a device as unmanaged and will not configure it.                                                                                                                                                                                                                       |
| ROOT CAUSE  | NetworkManager has a list of unmanaged devices specified in its config; matching devices are ignored.                                                                                                                                                                                         |
| CAUSE       | Interface explicitly unmanaged in /etc/NetworkManager/conf.d/, or keyfile missing.                                                                                                                                                                                                            |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check: nmcli device status</li><li>• Edit unmanaged list: /etc/NetworkManager/conf.d/99-unmanaged.conf</li><li>• Set managed: nmcli device set &lt;iface&gt; managed yes</li><li>• Restart NetworkManager: systemctl restart NetworkManager</li></ul> |

**#108 DHCP: No lease obtained**

|             |                                                                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A DHCP client could not obtain an IP address lease from a DHCP server.                                                                                                                                                                                                                           |
| ROOT CAUSE  | DHCP DISCOVER was broadcast but no DHCP OFFER was received within the timeout.                                                                                                                                                                                                                   |
| CAUSE       | No DHCP server on network, server pool exhausted, firewall blocking UDP 67/68, or MAC filtered.                                                                                                                                                                                                  |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check: dhclient -v &lt;interface&gt; for verbose output</li><li>• Verify DHCP server reachability and pool</li><li>• Check firewall on DHCP server for UDP 67</li><li>• Verify MAC not on exclusion list</li><li>• Use static IP as workaround</li></ul> |

**#109 BGP: hold timer expired**

|             |                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A BGP session went down because the hold timer expired without receiving a KEEPALIVE or UPDATE.                                                                                                                                                                                                |
| ROOT CAUSE  | BGP peers exchange KEEPALIVE messages; if none arrive within the hold time, the session is torn down.                                                                                                                                                                                          |
| CAUSE       | Network outage, high CPU on router, authentication mismatch, or firewall blocking BGP port 179.                                                                                                                                                                                                |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check BGP session: vtysh -c "show bgp summary"</li><li>• Verify TCP 179 connectivity between peers</li><li>• Check authentication (MD5 password) match</li><li>• Review router CPU: top in router shell</li><li>• Tune keepalive/hold timers</li></ul> |

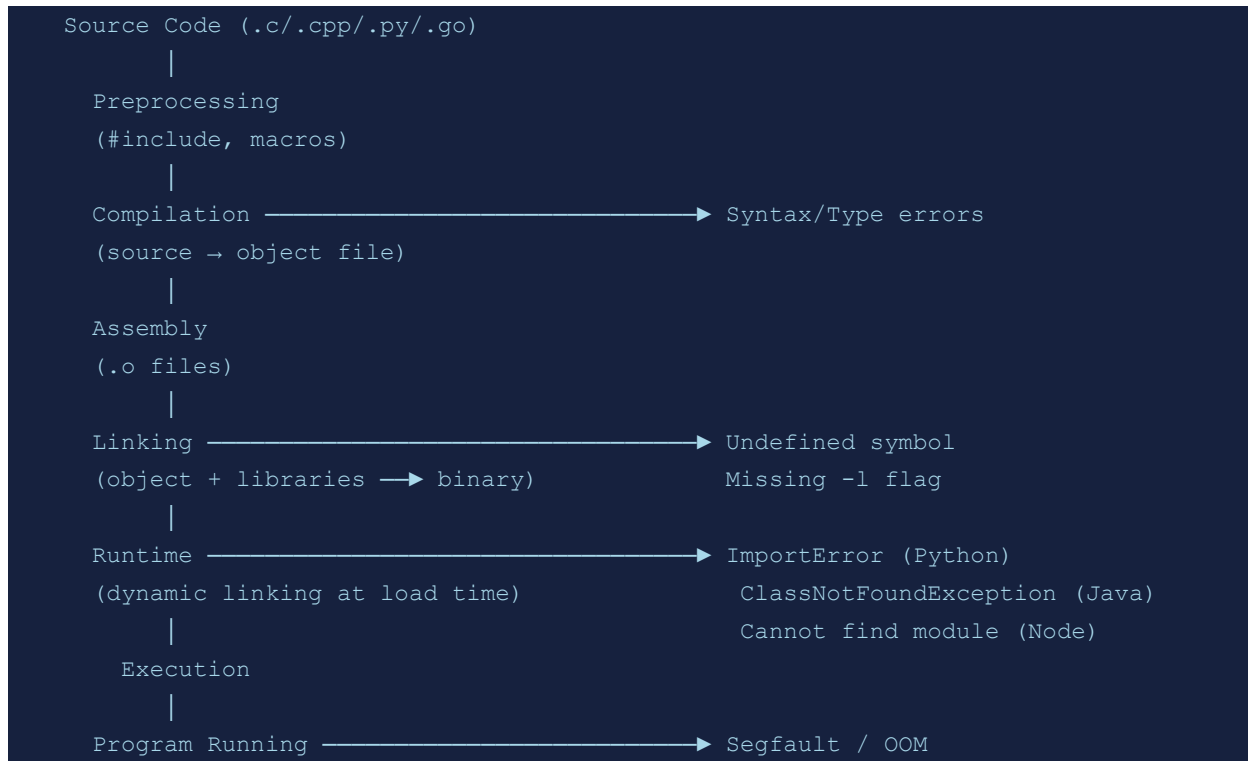
**#110 wireguard: Handshake did not complete**

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| DESCRIPTION | WireGuard cannot establish an encrypted tunnel because the handshake timed out. |
|-------------|---------------------------------------------------------------------------------|

|                   |                                                                                                                                                                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ROOT CAUSE</b> | WireGuard requires UDP connectivity between peers and matching cryptographic keys for the handshake to succeed.                                                                                                                                                                                                  |
| <b>CAUSE</b>      | Wrong peer public key, incorrect endpoint IP/port, firewall blocking UDP, or NAT issue.                                                                                                                                                                                                                          |
| <b>SOLUTION</b>   | <ul style="list-style-type: none"><li>• Check: wg show and verify keys</li><li>• Test UDP reachability to peer endpoint</li><li>• Check firewall allows WireGuard UDP port</li><li>• Verify AllowedIPs match traffic</li><li>• Enable persistent keepalive for NAT traversal: PersistentKeepalive = 25</li></ul> |

## Section 12: Programming & Build Errors (111–120)

**Diagram: Compilation & Build Error Flow**



### #111 gcc: undefined reference to symbol

|                    |                                                                                                                                                                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | The linker cannot resolve a symbol referenced in the object files during compilation.                                                                                                                                                                                                                    |
| <b>ROOT CAUSE</b>  | The linker searches specified libraries for symbols; if a library providing the symbol is not linked, the reference remains unresolved.                                                                                                                                                                  |
| <b>CAUSE</b>       | Missing -l flag, wrong library order, or library not installed.                                                                                                                                                                                                                                          |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>Add missing library: <code>gcc ... -lpthread -lm</code></li> <li>Check library order (dependencies after dependents)</li> <li>Find library: <code>ldconfig -p   grep libname</code></li> <li>Install dev package: <code>apt install libname-dev</code></li> </ul> |

### #112 cmake: Could NOT find package

|                    |                                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | CMake cannot locate a required dependency package during the configuration phase.                                                            |
| <b>ROOT CAUSE</b>  | CMake's <code>find_package()</code> searches standard and custom paths; if the package is not installed or not in the search path, it fails. |
| <b>CAUSE</b>       | Package not installed, dev headers missing, or non-standard install prefix.                                                                  |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>Install dev package: <code>apt install lib&lt;name&gt;-dev</code></li> </ul>                          |



- Set hint: `cmake -D<Name>_ROOT=/opt/mylib ..`
- Install to standard prefix
- Check CMakeLists.txt for exact package name required

### #113 python: ImportError: No module named

|             |                                                                                                                                                                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | Python cannot import a module because it is not installed or not in the Python path.                                                                                                                                                                                                                                                               |
| ROOT CAUSE  | Python's import system searches sys.path; if the module directory is not there, import fails.                                                                                                                                                                                                                                                      |
| CAUSE       | Module not installed, wrong virtual environment, or PYTHONPATH not set.                                                                                                                                                                                                                                                                            |
| SOLUTION    | <ul style="list-style-type: none"><li>• Install: <code>pip install &lt;module&gt;</code></li><li>• Check venv: <code>which python</code> and <code>pip list</code></li><li>• Add to path: <code>export PYTHONPATH=/path/to/module:\$PYTHONPATH</code></li><li>• Use full path to python: <code>/path/to/venv/bin/python script.py</code></li></ul> |

### #114 java: ClassNotFoundException

|             |                                                                                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | Java cannot find the specified class at runtime, causing the JVM to throw an exception.                                                                                                                                                                                                                 |
| ROOT CAUSE  | The Java ClassLoader searches the classpath; if the class's JAR or directory is absent, loading fails.                                                                                                                                                                                                  |
| CAUSE       | JAR not in classpath, wrong package name, or conflicting class versions.                                                                                                                                                                                                                                |
| SOLUTION    | <ul style="list-style-type: none"><li>• Set classpath: <code>java -cp /path/to/jar:. MainClass</code></li><li>• Or use: <code>export CLASSPATH=/path/to/jar:\$CLASSPATH</code></li><li>• Check dependency in build tool (Maven/Gradle)</li><li>• Verify class name spelling including package</li></ul> |

### #115 go build: cannot find package

|             |                                                                                                                                                                                                                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The Go toolchain cannot locate a package required for building.                                                                                                                                                                                                                                                               |
| ROOT CAUSE  | Go modules resolve package paths via go.mod and the module proxy; missing or unreachable packages fail resolution.                                                                                                                                                                                                            |
| CAUSE       | Package not in go.mod, GOPROXY unreachable, or GOPATH issue in GOPATH mode.                                                                                                                                                                                                                                                   |
| SOLUTION    | <ul style="list-style-type: none"><li>• Add dependency: <code>go get &lt;package@version&gt;</code></li><li>• Update modules: <code>go mod tidy</code></li><li>• Set proxy: <code>GOPROXY=https://proxy.golang.org go build</code></li><li>• Use vendor: <code>go mod vendor &amp;&amp; go build -mod=vendor</code></li></ul> |

### #116 make: No rule to make target

|                    |                                                                                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | GNU make cannot find a rule to build the specified target.                                                                                                                                                                                                                  |
| <b>ROOT CAUSE</b>  | make searches the Makefile for a rule matching the target; if absent and no implicit rule applies, it fails.                                                                                                                                                                |
| <b>CAUSE</b>       | Typo in target name, missing Makefile, or target not defined.                                                                                                                                                                                                               |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• List available targets: make help or grep "[a-z]" Makefile</li><li>• Check spelling of target</li><li>• Ensure you are in the correct directory with Makefile</li><li>• Generate Makefile if using autotools: ./configure</li></ul> |

### #117 ruby: Gem not found

|                    |                                                                                                                                                                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A Ruby gem required by the application is not installed or not available in the load path.                                                                                                                                                                    |
| <b>ROOT CAUSE</b>  | Bundler and Ruby's require mechanism search GEM_PATH for the gem; absence causes LoadError.                                                                                                                                                                   |
| <b>CAUSE</b>       | gem not installed, wrong Ruby version, or bundle install not run.                                                                                                                                                                                             |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Install gem: gem install &lt;gemname&gt;</li><li>• With Bundler: bundle install</li><li>• Check Ruby version: ruby -v and rbenv/rvm status</li><li>• Use bundle exec &lt;command&gt; to use correct gem set</li></ul> |

### #118 node: Cannot find module

|                    |                                                                                                                                                                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | Node.js cannot resolve and load the specified module during require/import.                                                                                                                                                                                                        |
| <b>ROOT CAUSE</b>  | Node's module resolution algorithm searches node_modules directories up the directory tree; missing modules cause failure.                                                                                                                                                         |
| <b>CAUSE</b>       | npm/yarn install not run, node_modules deleted, or wrong module name.                                                                                                                                                                                                              |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Install dependencies: npm install or yarn</li><li>• Check node_modules: ls node_modules/&lt;module&gt;</li><li>• Reinstall: rm -rf node_modules &amp;&amp; npm install</li><li>• Verify module name in package.json dependencies</li></ul> |

### #119 bash: command not found

|                    |                                                                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | The shell cannot find the specified command in any directory listed in PATH.                                                                                                                                                                   |
| <b>ROOT CAUSE</b>  | The shell searches each directory in PATH for an executable matching the command name.                                                                                                                                                         |
| <b>CAUSE</b>       | Package not installed, PATH not including install directory, or typo.                                                                                                                                                                          |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Check PATH: echo \$PATH</li><li>• Find command: which &lt;command&gt; or type &lt;command&gt;</li><li>• Install package providing command</li><li>• Add to PATH: export PATH=\$PATH:/new/dir</li></ul> |

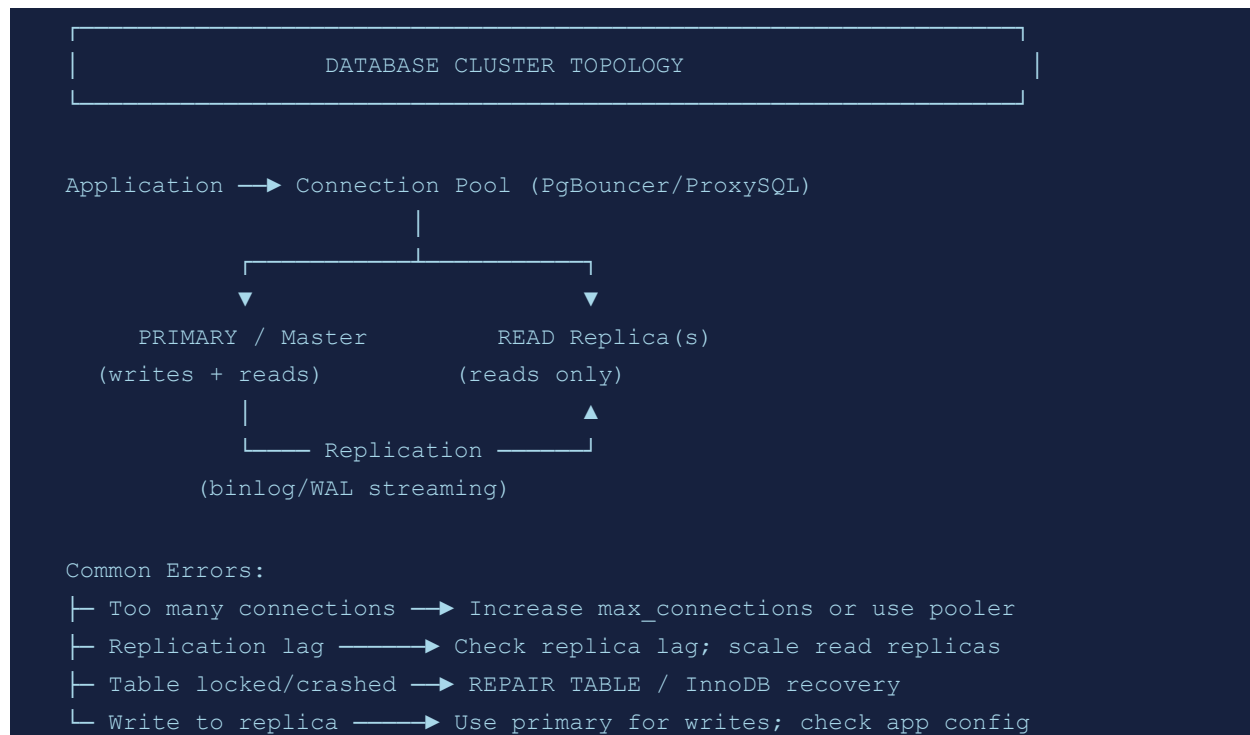
- Check if command needs execution permission: `chmod +x /path/to/cmd`

**#120 Ansible: UNREACHABLE!**

|                    |                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | Ansible cannot connect to a managed host to execute tasks.                                                                                                                                                                                                                                                                                                           |
| <b>ROOT CAUSE</b>  | Ansible uses SSH (by default) to connect to targets; if the connection fails, all tasks are marked unreachable.                                                                                                                                                                                                                                                      |
| <b>CAUSE</b>       | SSH not running on target, wrong credentials, firewall blocking, or inventory misconfiguration.                                                                                                                                                                                                                                                                      |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Test manually: <code>ssh user@target_host</code></li><li>• Check SSH service on target: <code>systemctl status sshd</code></li><li>• Verify inventory: <code>ansible-inventory --list</code></li><li>• Test ping: <code>ansible &lt;host&gt; -m ping -vvv</code></li><li>• Check become settings if sudo is needed</li></ul> |

## Section 13: Database Errors (121–130)

### Diagram: Database Connection & Replication Topology



## #121 MySQL: Too many connections

|             |                                                                                                                                                                                                                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | MySQL refuses new connections because the maximum connection count is reached.                                                                                                                                                                                                                                              |
| ROOT CAUSE  | MySQL maintains a global connection pool limited by max_connections; exceeding it returns error 1040.                                                                                                                                                                                                                       |
| CAUSE       | Application connection leak, connection pool misconfigured, or max_connections too low.                                                                                                                                                                                                                                     |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check: SHOW STATUS LIKE "Threads_connected"</li><li>• Increase: SET GLOBAL max_connections = 500</li><li>• Set permanently in my.cnf: max_connections=500</li><li>• Implement connection pooling (ProxySQL, PgBouncer)</li><li>• Fix connection leaks in application code</li></ul> |

```
#122 PostgreSQL: FATAL: role does not exist
```

|             |                                                                                          |
|-------------|------------------------------------------------------------------------------------------|
| DESCRIPTION | A connection to PostgreSQL fails because the specified database role is not defined.     |
| ROOT CAUSE  | PostgreSQL authentication maps connection user to pg_roles; an unknown role is rejected. |
| CAUSE       | Role not created, connecting as wrong user, or pg_hba.conf misconfiguration.             |

**SOLUTION**

- Create role: `CREATE ROLE <rolename> LOGIN;`
- Check roles: `\du` in `psql`
- Check `pg_hba.conf` allows connection for role
- Ensure application uses correct username

**#123 Redis: NOAUTH Authentication required**

|                    |                                                                                                                                                                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A Redis command fails because the client has not authenticated with the required password.                                                                                                                                                                                                              |
| <b>ROOT CAUSE</b>  | Redis <code>requirepass</code> enforces password authentication; unauthenticated clients receive NOAUTH error.                                                                                                                                                                                          |
| <b>CAUSE</b>       | Password set in <code>redis.conf</code> but client not sending AUTH command.                                                                                                                                                                                                                            |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Authenticate: <code>AUTH &lt;password&gt;</code></li><li>• Configure client with password in connection string</li><li>• Check <code>redis.conf</code> <code>requirepass</code> setting</li><li>• Use ACL for fine-grained access control in Redis 6+</li></ul> |

**#124 MongoDB: Not primary or slave**

|                    |                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A write operation to a MongoDB replica set member fails because it is not the primary.                                                                                                                                                                                                                                                          |
| <b>ROOT CAUSE</b>  | MongoDB replica sets enforce that writes go to the primary; secondaries return this error for writes.                                                                                                                                                                                                                                           |
| <b>CAUSE</b>       | Connecting to secondary directly, primary election in progress, or application not using replica set URI.                                                                                                                                                                                                                                       |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Use replica set connection string: <code>mongodb://host1,host2,host3/db?replicaSet=rs0</code></li><li>• Check primary: <code>rs.status()</code> in mongo shell</li><li>• Configure <code>readPreference</code> correctly for reads</li><li>• Wait for election to complete if primary changed</li></ul> |

**#125 sqlite: database is locked**

|                    |                                                                                                                                                                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | An SQLite operation fails because another process has a write lock on the database file.                                                                                                                                                                                                                                                                     |
| <b>ROOT CAUSE</b>  | SQLite uses file-level locking; concurrent writes are serialized, and a writer blocks other writers.                                                                                                                                                                                                                                                         |
| <b>CAUSE</b>       | Long-running transaction, application crash leaving lock, or concurrent processes using same DB file.                                                                                                                                                                                                                                                        |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Set busy timeout: <code>PRAGMA busy_timeout = 5000</code></li><li>• Check for processes: <code>fuser &lt;db_file&gt;</code></li><li>• Remove stale lock: <code>rm &lt;db_file&gt;.-journal</code> (only if no writers)</li><li>• Use WAL mode: <code>PRAGMA journal_mode=WAL</code> for better concurrency</li></ul> |

**#126 Elasticsearch: master\_not\_discovered\_exception**

|             |                                                                                                                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | An Elasticsearch node cannot discover a master node, causing cluster health to be RED.                                                                                                                                                                                                             |
| ROOT CAUSE  | Elasticsearch requires a quorum of master-eligible nodes to elect a master; if quorum is not met, the cluster is inoperative.                                                                                                                                                                      |
| CAUSE       | Not enough master-eligible nodes online, split-brain, or network partition.                                                                                                                                                                                                                        |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check cluster state: GET /_cluster/health</li><li>• Review discovery config: cluster.initial_master_nodes</li><li>• Verify minimum_master_nodes (ES 6): (N/2)+1</li><li>• Ensure network connectivity between nodes</li><li>• Restart down nodes</li></ul> |

**#127 MySQL: Table is crashed**

|             |                                                                                                                                                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A MySQL table is in a crashed state and cannot be read or written.                                                                                                                                                                                                                    |
| ROOT CAUSE  | The MyISAM storage engine (and sometimes InnoDB) can mark a table as crashed when it detects structural inconsistency.                                                                                                                                                                |
| CAUSE       | Power failure during write, disk error, or MySQL killed during table operation.                                                                                                                                                                                                       |
| SOLUTION    | <ul style="list-style-type: none"><li>• Repair: REPAIR TABLE &lt;table&gt;;</li><li>• Or: myisamchk -r /var/lib/mysql/&lt;db&gt;/&lt;table&gt;.MYI</li><li>• For InnoDB: set innodb_force_recovery=1 to 6 incrementally</li><li>• Restore from backup for severe corruption</li></ul> |

**#128 PostgreSQL: could not connect to server: No such file or directory**

|             |                                                                                                                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The psql client cannot connect to PostgreSQL because the UNIX socket file does not exist.                                                                                                                                                                       |
| ROOT CAUSE  | PostgreSQL creates a UNIX socket in /var/run/postgresql/ by default; if the server is not running, no socket exists.                                                                                                                                            |
| CAUSE       | PostgreSQL service not running, or socket directory path mismatch.                                                                                                                                                                                              |
| SOLUTION    | <ul style="list-style-type: none"><li>• Start service: systemctl start postgresql</li><li>• Check socket: ls /var/run/postgresql/</li><li>• Specify host explicitly: psql -h 127.0.0.1 -p 5432</li><li>• Check service logs: journalctl -u postgresql</li></ul> |

**#129 RabbitMQ: Channel closed with error**

|             |                                                                                      |
|-------------|--------------------------------------------------------------------------------------|
| DESCRIPTION | A RabbitMQ channel is closed by the broker with an error, interrupting message flow. |
|-------------|--------------------------------------------------------------------------------------|

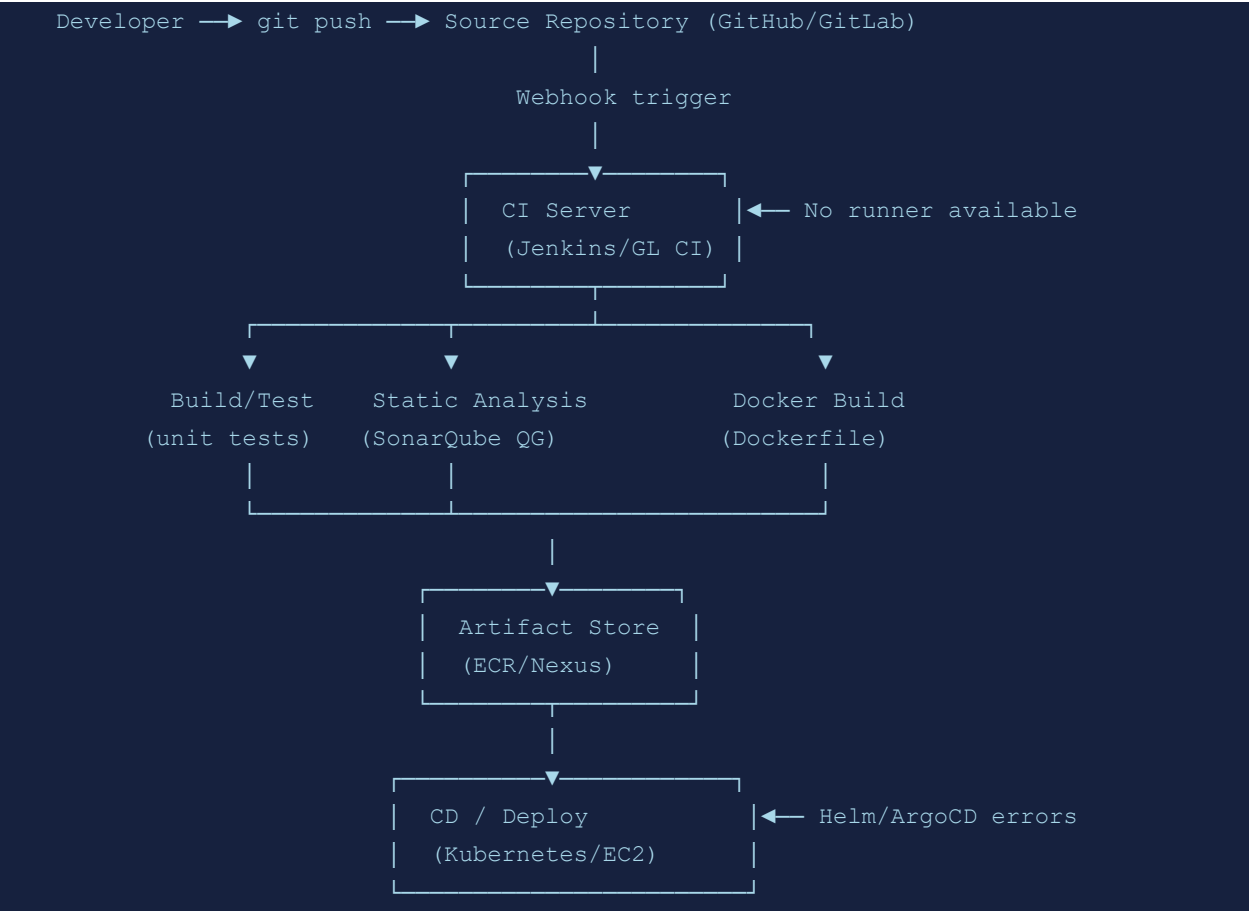
|                   |                                                                                                                                                                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ROOT CAUSE</b> | RabbitMQ closes channels when protocol violations occur, such as declaring a queue with wrong parameters or exceeding channel limits.                                                                                                                                                            |
| <b>CAUSE</b>      | Queue property mismatch (durable/exclusive), channel limit exceeded, or resource alarm.                                                                                                                                                                                                          |
| <b>SOLUTION</b>   | <ul style="list-style-type: none"><li>• Check RabbitMQ logs: /var/log/rabbitmq/</li><li>• Verify queue declaration parameters match existing queue</li><li>• Increase channel limit: channel_max in rabbitmq.conf</li><li>• Check memory/disk alarms: rabbitmqctl status   grep alarms</li></ul> |

### #130 Cassandra: WriteTimeoutException

|                    |                                                                                                                                                                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | A Cassandra write operation timed out because not enough replicas acknowledged the write within the timeout.                                                                                                                                                                                |
| <b>ROOT CAUSE</b>  | Cassandra requires a configurable number of replica acknowledgments (consistency level); if replicas are slow or down, the write times out.                                                                                                                                                 |
| <b>CAUSE</b>       | Node down, GC pause, overloaded cluster, or consistency level too strict.                                                                                                                                                                                                                   |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Check: nodetool status to see node health</li><li>• Lower consistency level temporarily: CONSISTENCY ONE</li><li>• Check GC: nodetool gcstats</li><li>• Review write latency: nodetool tpstats</li><li>• Scale cluster or tune compaction</li></ul> |

Section 14: CI/CD & Automation Errors (131–140)

Diagram: CI/CD Pipeline Flow & Failure Points



| #131 Jenkins: Build failed: executor not available |                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                                        | A Jenkins build cannot start because no executor is available to run it.                                                                                                                                                                                                                            |
| ROOT CAUSE                                         | Jenkins queues jobs when all executors on matching agents are occupied; jobs remain pending indefinitely if no agent becomes free.                                                                                                                                                                  |
| CAUSE                                              | All agents busy, agent offline, or label mismatch between job and available agents.                                                                                                                                                                                                                 |
| SOLUTION                                           | <ul style="list-style-type: none"><li>• Check agent status in Jenkins UI &gt; Manage Nodes</li><li>• Restart offline agents</li><li>• Add more agents or increase executor count</li><li>• Review job label requirements</li><li>• Use cloud agents (EC2, Kubernetes) for elastic scaling</li></ul> |

| #132 GitLab CI: Runner not available |                                                                                              |
|--------------------------------------|----------------------------------------------------------------------------------------------|
| DESCRIPTION                          | A GitLab CI pipeline job is stuck because no registered runner matches the job requirements. |



|                   |                                                                                                                                                                                                                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ROOT CAUSE</b> | GitLab CI dispatches jobs to runners matching the job's tags and executor type; no available match leaves the job pending.                                                                                                                                                                                      |
| <b>CAUSE</b>      | All runners busy, runner offline, or job tags not matching any runner.                                                                                                                                                                                                                                          |
| <b>SOLUTION</b>   | <ul style="list-style-type: none"> <li>• Check runner status: Settings &gt; CI/CD &gt; Runners</li> <li>• Register new runner: <code>gitlab-runner register</code></li> <li>• Ensure runner tags match job tags or use untagged runner</li> <li>• Restart runner: <code>gitlab-runner restart</code></li> </ul> |

**#133 Terraform: Error acquiring state lock**

|                    |                                                                                                                                                                                                                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | Terraform cannot acquire the state lock because another operation holds it.                                                                                                                                                                                                             |
| <b>ROOT CAUSE</b>  | Terraform uses state locking (DynamoDB for S3 backend) to prevent concurrent modifications; an existing lock blocks new operations.                                                                                                                                                     |
| <b>CAUSE</b>       | Previous Terraform run crashed without releasing lock, or concurrent apply.                                                                                                                                                                                                             |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Check lock info: <code>terraform force-unlock &lt;lock-id&gt;</code></li> <li>• Verify no concurrent run is active</li> <li>• Check DynamoDB table for stale lock entry</li> <li>• Use <code>-lock=false</code> only in emergencies</li> </ul> |

**#134 Ansible: Error in vault password**

|                    |                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | Ansible cannot decrypt an Ansible Vault-encrypted variable or file.                                                                                                                                                                                                                                                                                              |
| <b>ROOT CAUSE</b>  | Ansible Vault uses AES-256 encryption; the decryption requires the exact password used during encryption.                                                                                                                                                                                                                                                        |
| <b>CAUSE</b>       | Wrong vault password provided, different vault ID used, or encrypted file corrupted.                                                                                                                                                                                                                                                                             |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Specify password file: <code>ansible-playbook --vault-password-file=vault.pass</code></li> <li>• Or: <code>ansible-playbook --ask-vault-pass</code></li> <li>• Re-encrypt with correct password: <code>ansible-vault rekey vars.yml</code></li> <li>• Check <code>ANSIBLE_VAULT_PASSWORD_FILE</code> env var</li> </ul> |

**#135 Helm: chart not found in repository**

|                    |                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | Helm cannot locate the specified chart in the configured chart repository.                                                                                                                                                                                                                |
| <b>ROOT CAUSE</b>  | Helm searches local repository cache for the chart name and version; cache miss or wrong repo causes failure.                                                                                                                                                                             |
| <b>CAUSE</b>       | Repository not added, cache not updated, or chart name/version typo.                                                                                                                                                                                                                      |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"> <li>• Add repo: <code>helm repo add &lt;name&gt; &lt;url&gt;</code></li> <li>• Update cache: <code>helm repo update</code></li> <li>• Search: <code>helm search repo &lt;chart&gt;</code></li> <li>• Verify chart name and version spelling</li> </ul> |

**#136 kubectl: error from server (Forbidden)**

|             |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A kubectl command fails because the requesting user or service account lacks permission.                                                                                                                                                                                                                                                                            |
| ROOT CAUSE  | Kubernetes RBAC evaluates the request against ClusterRoles/Roles bound to the principal; no matching allow rule results in denial.                                                                                                                                                                                                                                  |
| CAUSE       | Missing RBAC binding, wrong namespace, or cluster-admin rights not granted.                                                                                                                                                                                                                                                                                         |
| SOLUTION    | <ul style="list-style-type: none"><li>• Check permissions: <code>kubectl auth can-i &lt;verb&gt; &lt;resource&gt; --as=&lt;user&gt;</code></li><li>• Create binding: <code>kubectl create clusterrolebinding</code></li><li>• Review existing bindings: <code>kubectl get rolebinding,clusterrolebinding -A</code></li><li>• Apply correct RBAC manifests</li></ul> |

**#137 GitHub Actions: Secret not found**

|             |                                                                                                                                                                                                                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A workflow step fails because a referenced secret is not defined in the repository or organization secrets.                                                                                                                                                                                                                       |
| ROOT CAUSE  | GitHub Actions injects secrets as environment variables only if they are defined at the appropriate scope; undefined secrets result in empty values.                                                                                                                                                                              |
| CAUSE       | Secret not added in Settings > Secrets, wrong name, or environment not configured.                                                                                                                                                                                                                                                |
| SOLUTION    | <ul style="list-style-type: none"><li>• Add secret: Settings &gt; Secrets and variables &gt; Actions</li><li>• Verify secret name matches workflow reference exactly (case-sensitive)</li><li>• Check environment secrets if using environments</li><li>• Use <code>\${{ secrets.SECRET_NAME }}</code> syntax correctly</li></ul> |

**#138 SonarQube: Quality Gate failed**

|             |                                                                                                                                                                                                                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A CI/CD pipeline build fails because SonarQube analysis found issues exceeding the Quality Gate thresholds.                                                                                                                                                                                                                               |
| ROOT CAUSE  | SonarQube Quality Gates define conditions (coverage, bugs, code smells) that must pass; failing conditions block the build.                                                                                                                                                                                                               |
| CAUSE       | New code introduces bugs/vulnerabilities, coverage dropped, or duplications increased.                                                                                                                                                                                                                                                    |
| SOLUTION    | <ul style="list-style-type: none"><li>• Review SonarQube project dashboard for specific failures</li><li>• Fix reported issues in code</li><li>• Adjust Quality Gate conditions if overly strict</li><li>• Use <code>sonar.qualitygate.wait=true</code> to block appropriately</li><li>• Add exclusions for test/generated code</li></ul> |

**#139 Packer: error waiting for SSH**

|             |                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------|
| DESCRIPTION | A Packer image build fails because it cannot SSH into the builder instance during provisioning. |
|-------------|-------------------------------------------------------------------------------------------------|

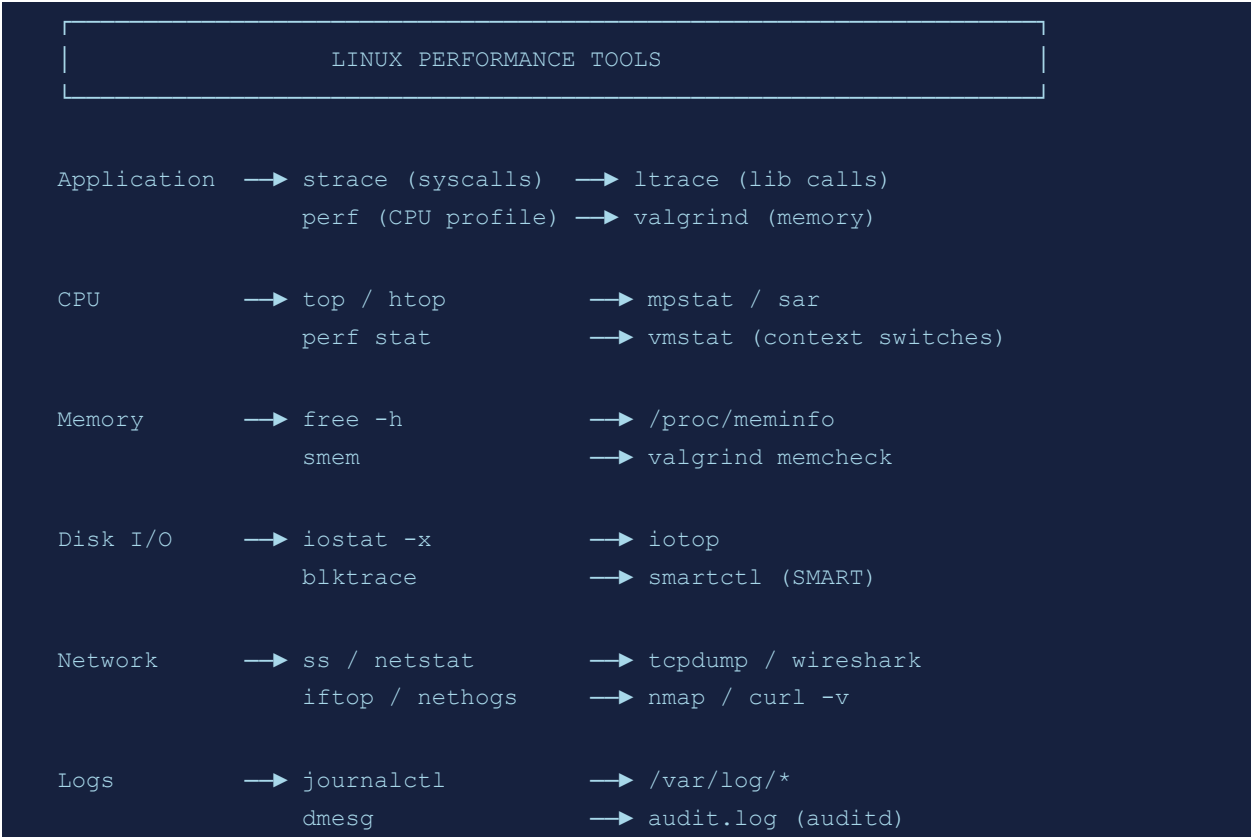
|                   |                                                                                                                                                                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ROOT CAUSE</b> | Packer starts a VM/cloud instance and waits for SSH to become available to run provisioners; timeout means SSH never became reachable.                                                                                                                                                                           |
| <b>CAUSE</b>      | Security group/firewall blocking SSH, wrong SSH key, instance boot failure, or timeout too short.                                                                                                                                                                                                                |
| <b>SOLUTION</b>   | <ul style="list-style-type: none"><li>• Check cloud security group allows SSH from Packer host</li><li>• Increase ssh_timeout in template</li><li>• Verify ssh_username matches AML user</li><li>• Check cloud instance console for boot errors</li><li>• Add debug logging: PACKER_LOG=1 packer build</li></ul> |

#### #140 ArgoCD: Application out of sync

|                    |                                                                                                                                                                                                                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | ArgoCD reports an application as OutOfSync because the live cluster state does not match the desired Git state.                                                                                                                                                                                                                       |
| <b>ROOT CAUSE</b>  | ArgoCD compares live Kubernetes resources with manifests in Git; any difference is flagged as out of sync.                                                                                                                                                                                                                            |
| <b>CAUSE</b>       | Manual changes to cluster resources, image tag updated in Git, or drift from kubectl apply.                                                                                                                                                                                                                                           |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Review diff: <code>argocd app diff &lt;app&gt;</code></li><li>• Sync: <code>argocd app sync &lt;app&gt;</code></li><li>• Enable auto-sync: <code>argocd app set &lt;app&gt; --sync-policy automated</code></li><li>• Fix manual drift and let ArgoCD manage all changes through Git</li></ul> |

Section 15: Performance & Observability (141–150)

Diagram: Linux Observability Tooling Map



| #141 perf: Permission denied (kernel.perf_event_paranoid) |                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION                                               | The perf profiling tool is denied access to kernel performance counters.                                                                                                                                                                                                                                                 |
| ROOT CAUSE                                                | The kernel parameter perf_event_paranoid controls access to perf events; high values restrict non-root profiling.                                                                                                                                                                                                        |
| CAUSE                                                     | Default paranoid setting restricts perf to root only.                                                                                                                                                                                                                                                                    |
| SOLUTION                                                  | <ul style="list-style-type: none"><li>Temporarily allow: <code>sysctl -w kernel.perf_event_paranoid=1</code></li><li>Set permanently in <code>/etc/sysctl.conf</code></li><li>Or run as root: <code>sudo perf stat &lt;command&gt;</code></li><li>Use <code>-1</code> to allow full access (security tradeoff)</li></ul> |

| #142 strace: attach: ptrace(PTRACE_SEIZE): Operation not permitted |                                                                                                        |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| DESCRIPTION                                                        | strace cannot attach to a running process to trace its system calls.                                   |
| ROOT CAUSE                                                         | ptrace requires either root or matching UID AND the process must not have a no_ptrace LSM restriction. |

|          |                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAUSE    | Process owned by different user, Yama ptrace_scope set to 1+, or LSM restriction.                                                                                                                                                                                                               |
| SOLUTION | <ul style="list-style-type: none"><li>• Run as root: <code>sudo strace -p &lt;PID&gt;</code></li><li>• Reduce Yama scope temporarily: <code>sysctl kernel.yama.ptrace_scope=0</code></li><li>• Or use same UID as target process</li><li>• Check AppArmor/SELinux ptrace restrictions</li></ul> |

#### #143 Prometheus: many-to-many matching not allowed

|             |                                                                                                                                                                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A PromQL query fails because a binary operation would result in a many-to-many label match.                                                                                                                                                                                                         |
| ROOT CAUSE  | PromQL binary operations require one-to-one or many-to-one matching; many-to-many is ambiguous and disallowed.                                                                                                                                                                                      |
| CAUSE       | Missing group_left/group_right modifier, or wrong label set in on() clause.                                                                                                                                                                                                                         |
| SOLUTION    | <ul style="list-style-type: none"><li>• Add group_left: <code>metric1 * on(instance) group_left(job) metric2</code></li><li>• Specify matching labels explicitly with on()</li><li>• Review cardinality of both sides of operation</li><li>• Use without() to exclude non-matching labels</li></ul> |

#### #144 Grafana: No data source found

|             |                                                                                                                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | A Grafana dashboard panel cannot load because the configured data source is missing or unreachable.                                                                                                                                                                        |
| ROOT CAUSE  | Grafana queries data sources by name or UID; a missing or misconfigured data source returns no data or an error.                                                                                                                                                           |
| CAUSE       | Data source deleted, Prometheus/InfluxDB unreachable, or dashboard imported without matching data source.                                                                                                                                                                  |
| SOLUTION    | <ul style="list-style-type: none"><li>• Add data source: Configuration &gt; Data Sources</li><li>• Verify data source URL is reachable from Grafana</li><li>• Re-link panels to correct data source</li><li>• Export/import dashboard with data source variables</li></ul> |

#### #145 tcpdump: no suitable device found

|             |                                                                                                                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | tcpdump cannot start because it cannot find or open the specified (or default) network interface.                                                                                                                 |
| ROOT CAUSE  | tcpdump requires a capture-capable interface; if the name is wrong or libpcap cannot open it, capture fails.                                                                                                      |
| CAUSE       | Wrong interface name, interface down, insufficient permissions, or no interfaces available.                                                                                                                       |
| SOLUTION    | <ul style="list-style-type: none"><li>• List interfaces: <code>tcpdump -D</code></li><li>• Specify correct interface: <code>tcpdump -i eth0</code></li><li>• Run as root or with CAP_NET_RAW capability</li></ul> |

- Bring up interface: `ip link set <iface> up`

**#146 valgrind: Mismatched free/delete/delete[]**

|             |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | Valgrind reports a memory error where memory was allocated with one allocator but freed with another.                                                                                                                                                                                                                                                               |
| ROOT CAUSE  | Valgrind's memcheck tracks allocations and their type; freeing with a mismatched function is undefined behavior.                                                                                                                                                                                                                                                    |
| CAUSE       | Using <code>free()</code> on <code>new[]</code> allocated memory, or <code>delete</code> on <code>malloc()</code> allocated memory.                                                                                                                                                                                                                                 |
| SOLUTION    | <ul style="list-style-type: none"> <li>• Match allocation and deallocation: <code>malloc/free</code>, <code>new/delete</code>, <code>new[]/delete[]</code></li> <li>• Review code for mixed C/C++ allocation styles</li> <li>• Run valgrind with <code>--leak-check=full</code> for full report</li> <li>• Fix all mismatches to prevent heap corruption</li> </ul> |

**#147 sar: Cannot open /var/log/sa/saXX**

|             |                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | The sar command cannot find historical data for the requested day.                                                                                                                                                                                                                                                      |
| ROOT CAUSE  | sar reads from daily sa files written by the sadc collector; if collection was not running or data rotated, files are absent.                                                                                                                                                                                           |
| CAUSE       | sysstat service not running, data not collected on that day, or rotation deleted files.                                                                                                                                                                                                                                 |
| SOLUTION    | <ul style="list-style-type: none"> <li>• Enable collection: <code>systemctl enable --now sysstat</code></li> <li>• Check service: <code>systemctl status sysstat</code></li> <li>• View available files: <code>ls /var/log/sa/</code></li> <li>• Adjust HISTORY setting in <code>/etc/sysstat/sysstat</code></li> </ul> |

**#148 gdb: No symbol table loaded**

|             |                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESCRIPTION | gdb cannot display function names or variable information because the binary lacks debug symbols.                                                                                                                                                                                                                                                                                          |
| ROOT CAUSE  | Debug symbols (DWARF data) map binary addresses to source code; stripped binaries have no symbol table.                                                                                                                                                                                                                                                                                    |
| CAUSE       | Binary compiled without <code>-g</code> flag, or <code>strip</code> run after compilation.                                                                                                                                                                                                                                                                                                 |
| SOLUTION    | <ul style="list-style-type: none"> <li>• Recompile with debug symbols: <code>gcc -g -O0 -o binary source.c</code></li> <li>• Or install debug package: <code>apt install &lt;package&gt;-dbg</code></li> <li>• Use debuginfod: <code>export DEBUGINFOD_URLS=https://debuginfod.elfutils.org/</code></li> <li>• Load symbols manually: <code>symbol-file /path/to/symbols</code></li> </ul> |

**#149 iostat: device not found in /proc/diskstats**

|             |                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------|
| DESCRIPTION | iostat cannot report statistics for a specified device because it is not in the kernel's diskstats. |
|-------------|-----------------------------------------------------------------------------------------------------|

|                   |                                                                                                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ROOT CAUSE</b> | iostat reads /proc/diskstats which lists block devices the kernel knows about; unknown names produce no data.                                                                                                                                                       |
| <b>CAUSE</b>      | Wrong device name, device not mounted, or device mapper name not used.                                                                                                                                                                                              |
| <b>SOLUTION</b>   | <ul style="list-style-type: none"><li>• List devices: lsblk or cat /proc/diskstats</li><li>• Use correct device name (e.g., dm-0 for LVM)</li><li>• Verify device is accessible: ls /dev/&lt;device&gt;</li><li>• For dm devices, map to name: dmsetup ls</li></ul> |

### #150 systemd-analyze: Permission denied

|                    |                                                                                                                                                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DESCRIPTION</b> | systemd-analyze cannot read unit file details due to insufficient permissions.                                                                                                                                                                                                                |
| <b>ROOT CAUSE</b>  | Some systemd-analyze operations require reading protected unit files or kernel interfaces that are restricted to root.                                                                                                                                                                        |
| <b>CAUSE</b>       | Running as non-root user for a system-level analysis operation.                                                                                                                                                                                                                               |
| <b>SOLUTION</b>    | <ul style="list-style-type: none"><li>• Run as root: sudo systemd-analyze blame</li><li>• Or: sudo systemd-analyze critical-chain</li><li>• For user units, no sudo needed: systemd-analyze --user blame</li><li>• Install polkit to allow specific unprivileged analyze operations</li></ul> |

# DevOps Shack

For updates, contributions, and feedback:

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